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(54) **CERVICAL DISTRACTION/IMPLANT DELIVERY DEVICE**

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(58) **Field of Classification Search**

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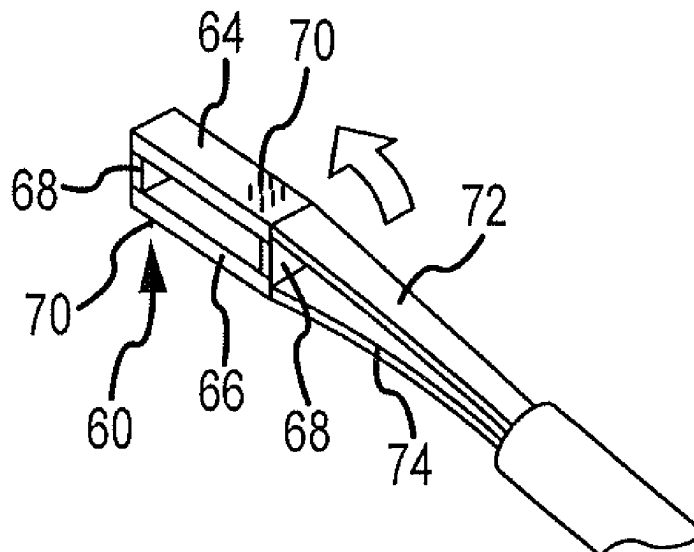
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(57) **ABSTRACT**

Systems for distracting a facet joint and positioning a permanent implant in the joint are disclosed. The implants serve to retain a distracted position of the facet joint which is achieved with positioning of the leading end of a distraction tool in the facet joint and then distracting or enlarging the joint a desired amount. The permanent implant could be part of the distraction mechanism which can be separated from the delivery tool once the joint has been distracted or an auxiliary implant may be positioned before the distraction mechanism is removed from the distracted joint. The permanent implants can be solid, mechanical devices that may have fixation means thereon to hold them in place or injected fluids such as hydrogels or fluids confined within balloons.

15 Claims, 23 Drawing Sheets



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continuation of application No. 15/488,989, filed on Apr. 17, 2017, now Pat. No. 10,238,501, which is a continuation of application No. 13/949,042, filed on Jul. 23, 2013, now Pat. No. 9,622,874, which is a continuation of application No. 12/559,193, filed on Sep. 14, 2009, now Pat. No. 8,512,347, which is a continuation-in-part of application No. 12/455,814, filed on Jun. 5, 2009, now Pat. No. 8,361,152, which is a continuation-in-part of application No. 12/317,682, filed on Dec. 23, 2008, now Pat. No. 8,267,966.

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* cited by examiner

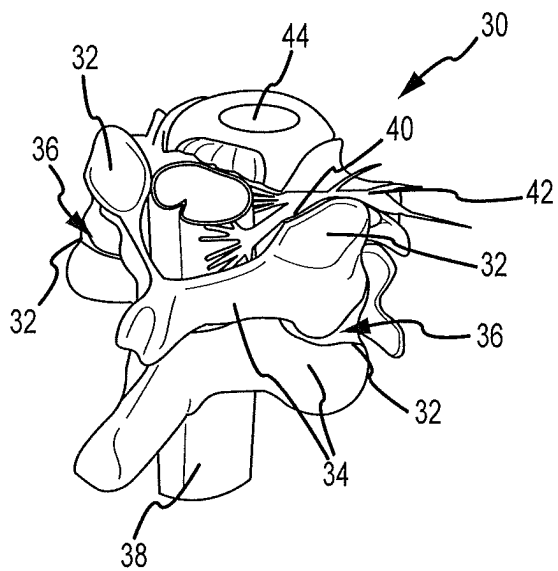


FIG. 1

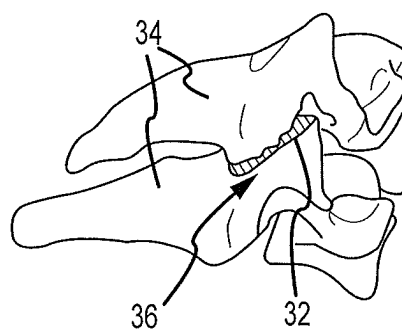


FIG. 2

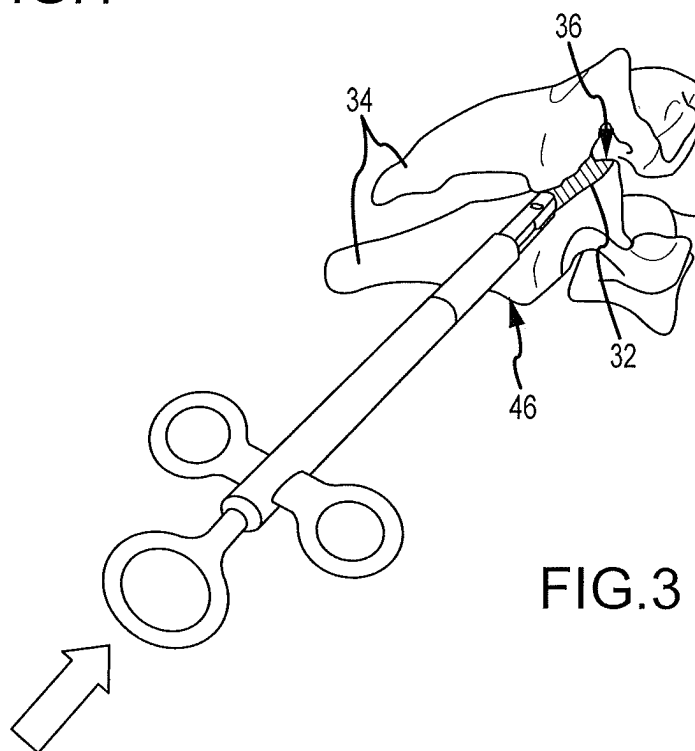


FIG. 3

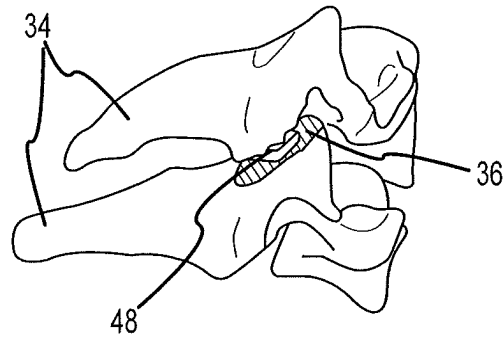


FIG. 4

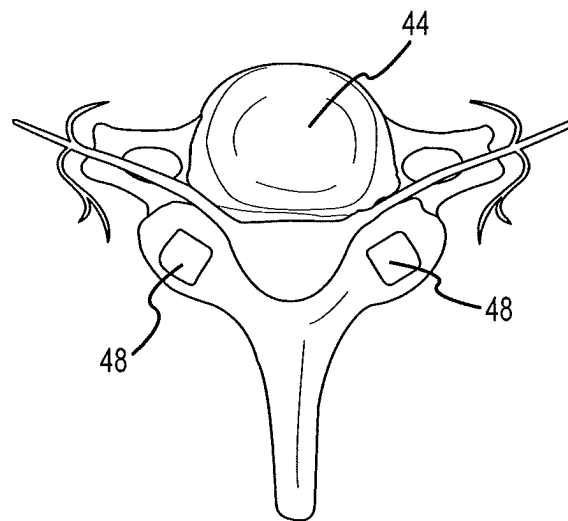
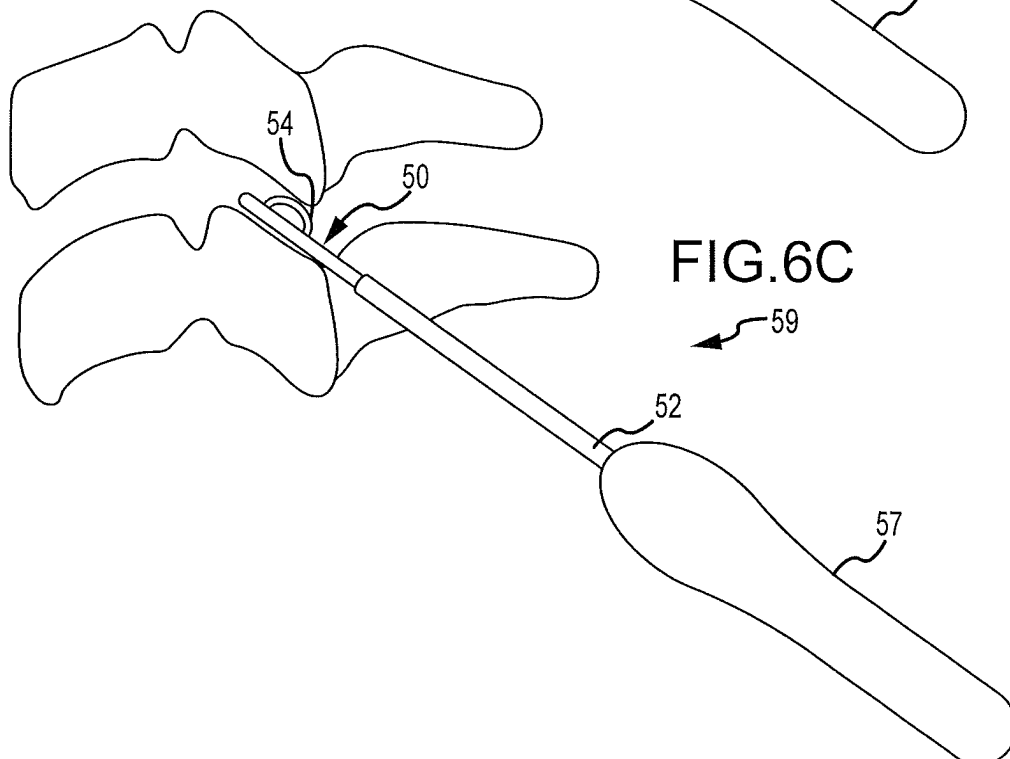
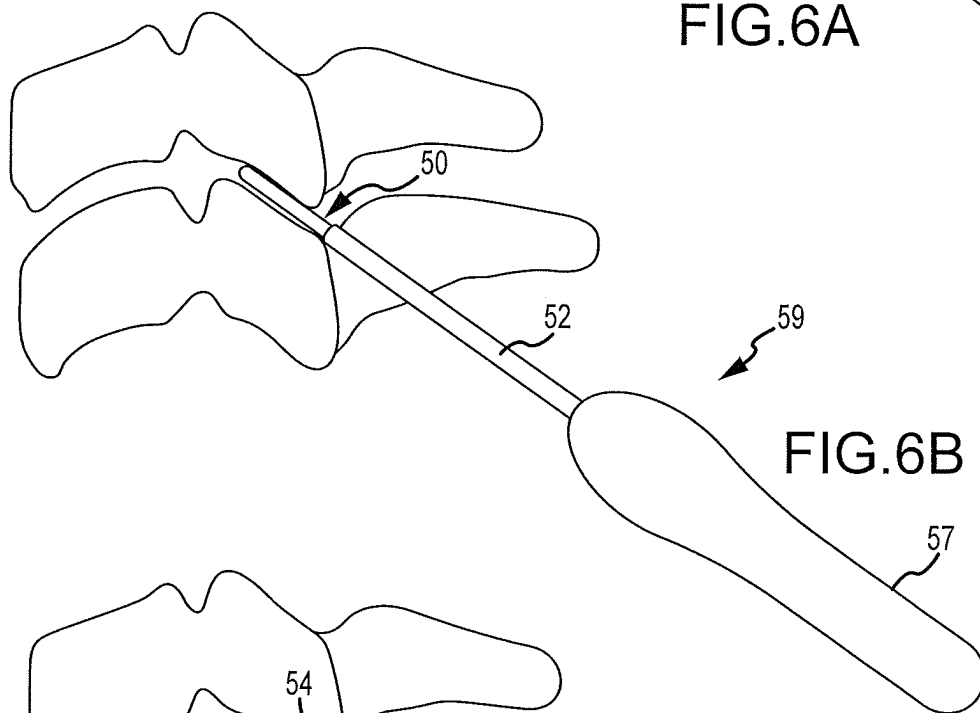
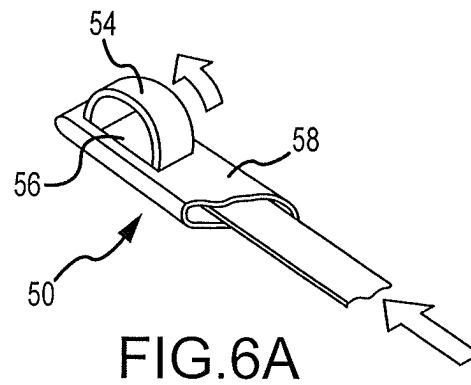


FIG. 5



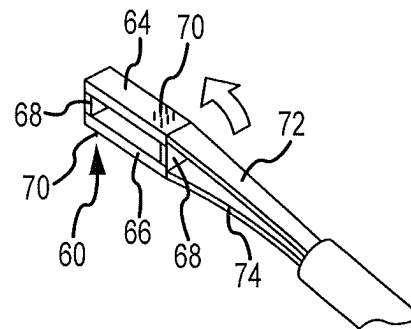


FIG. 7A

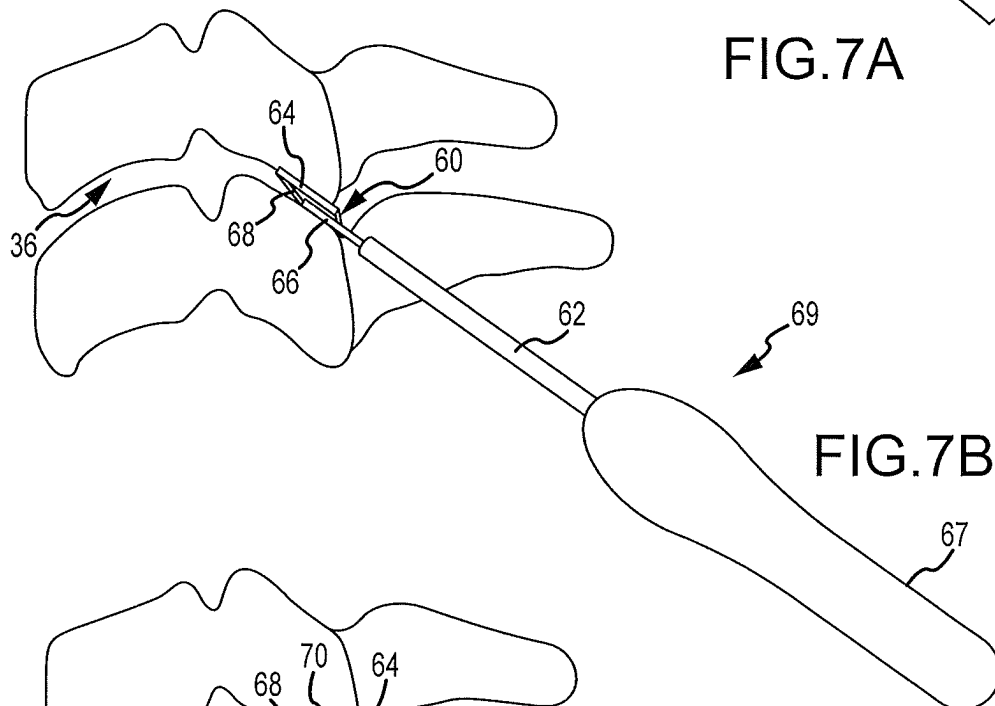


FIG. 7B

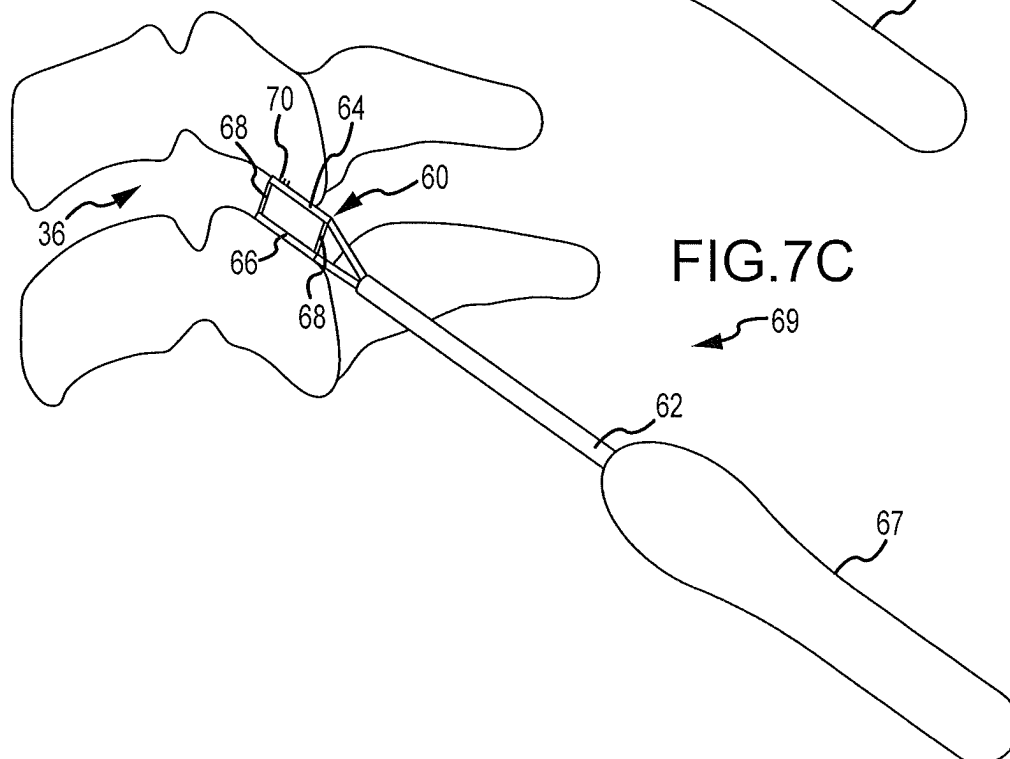


FIG. 7C

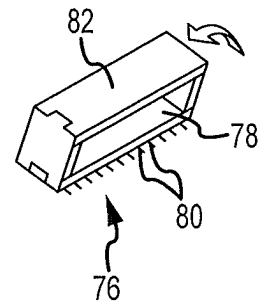


FIG. 8A

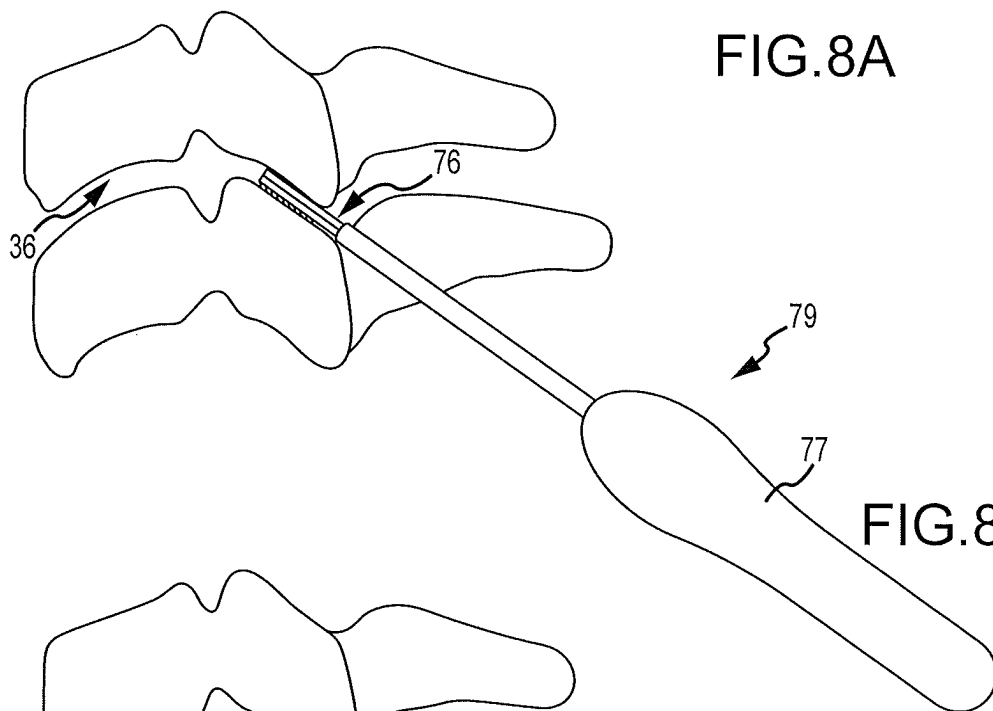


FIG. 8B

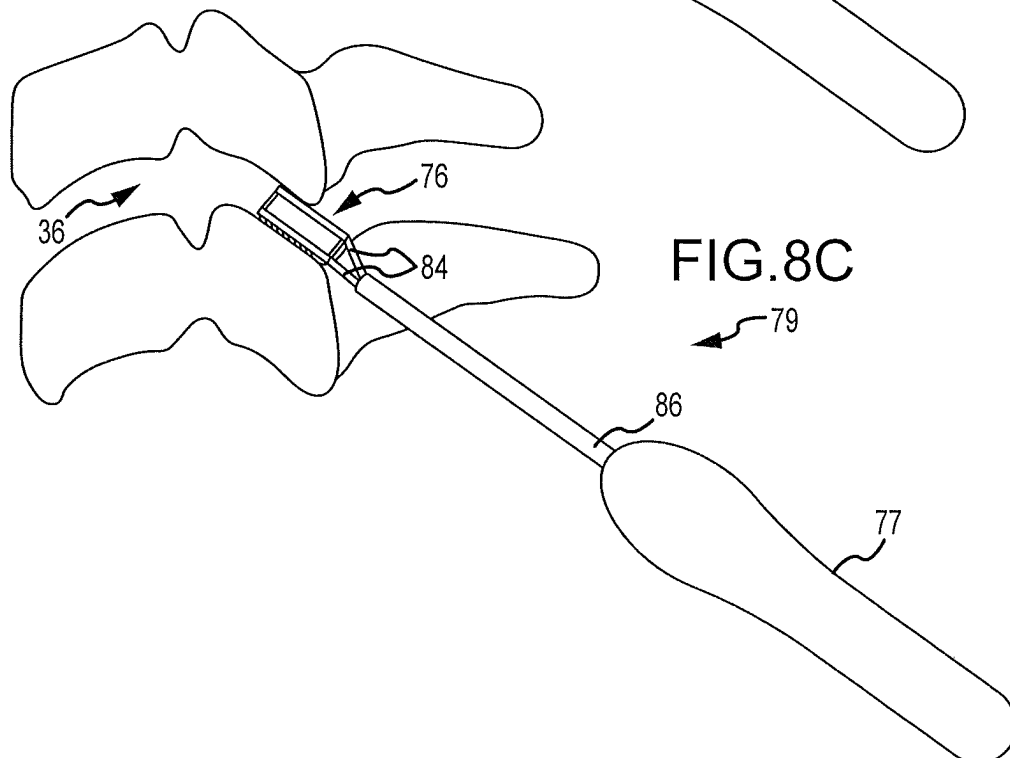


FIG. 8C

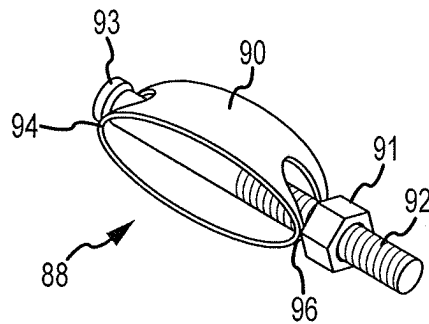


FIG. 9A

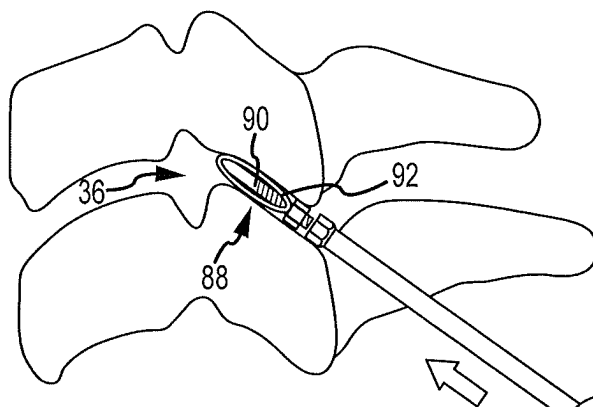


FIG. 9B

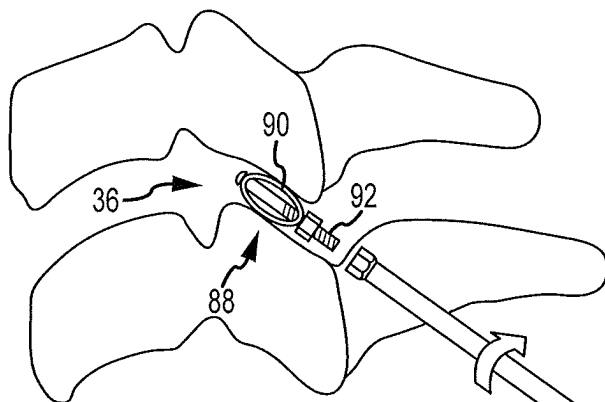


FIG. 9C

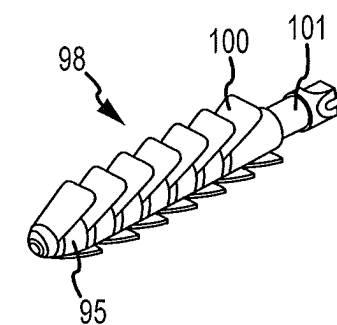


FIG. 10A

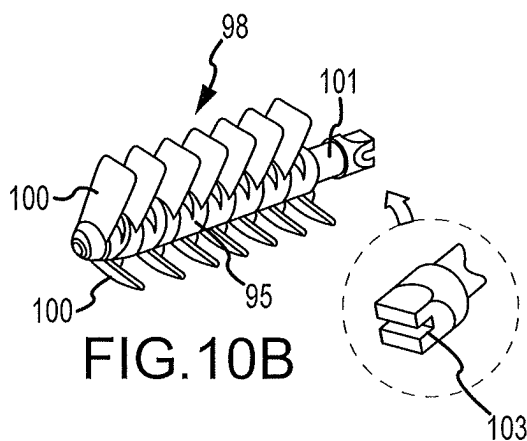


FIG. 10B

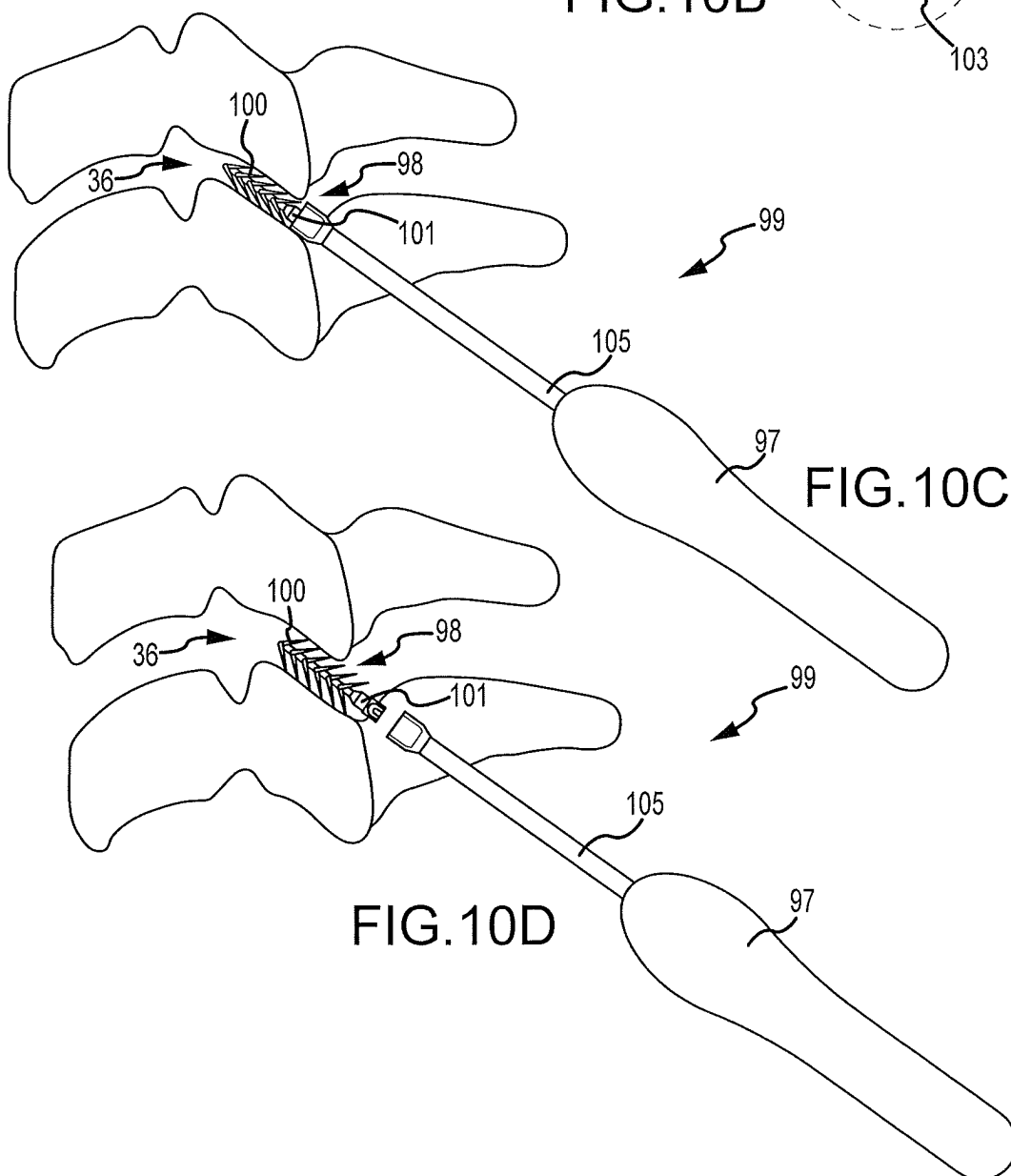


FIG. 10C

FIG. 10D

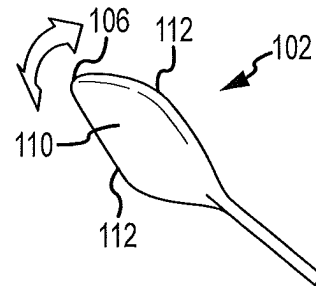


FIG. 11A

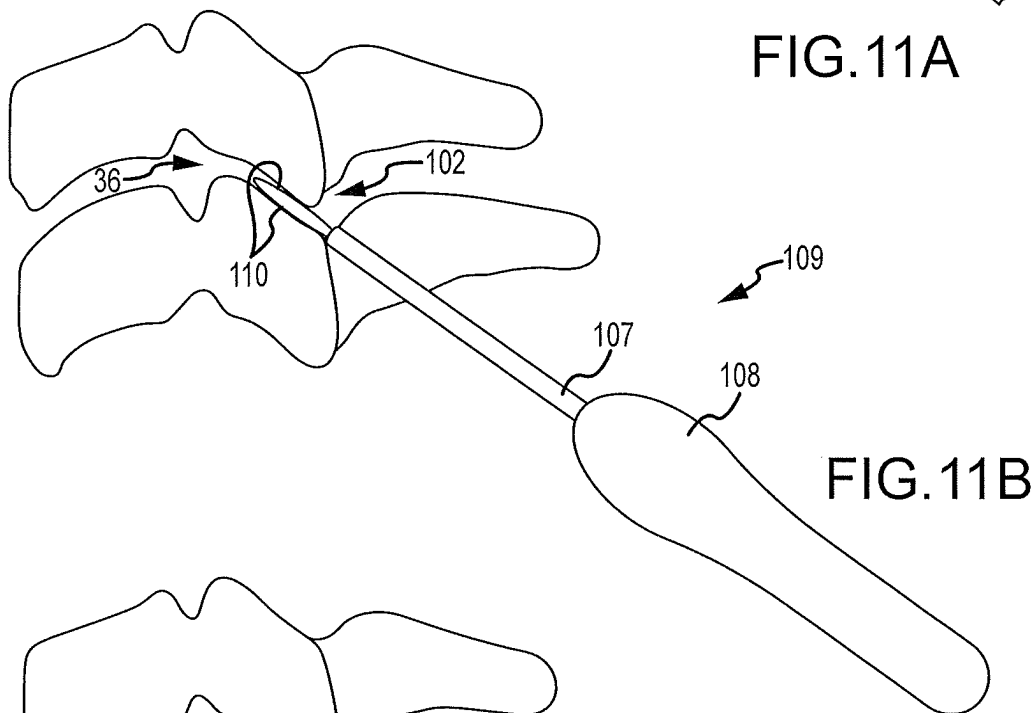


FIG. 11B

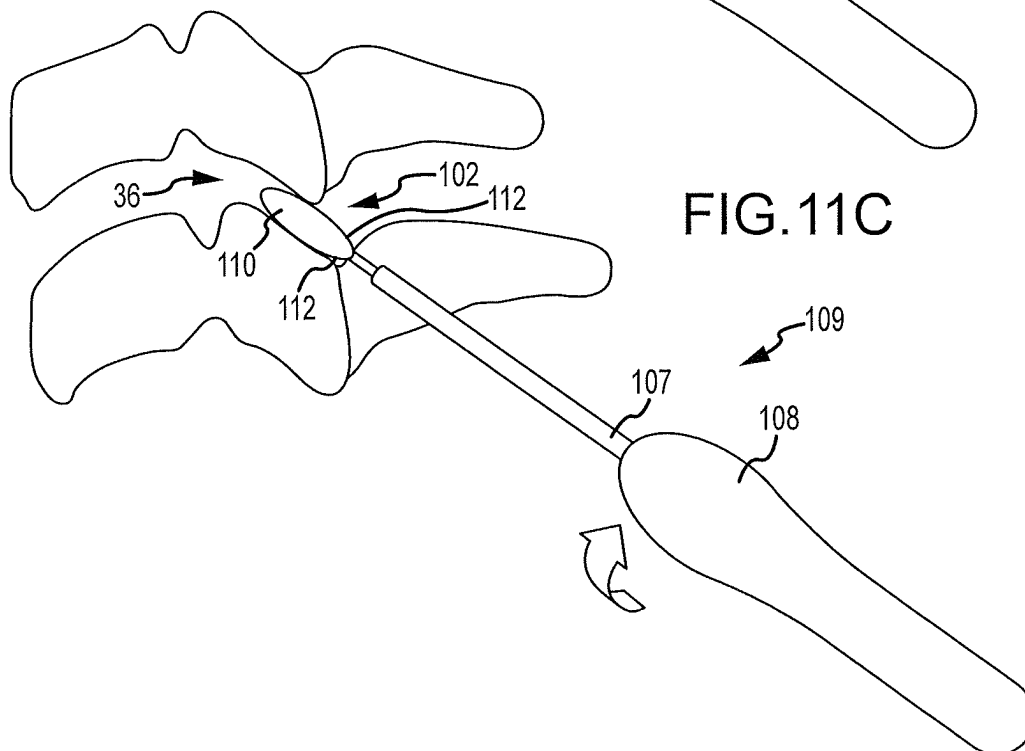


FIG. 11C

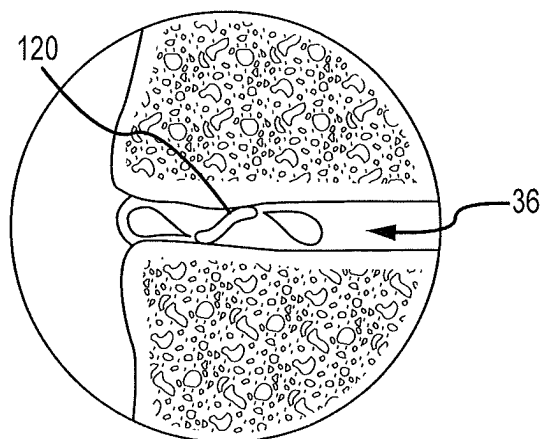
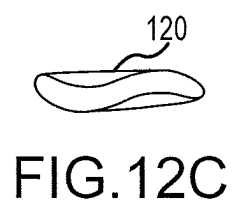
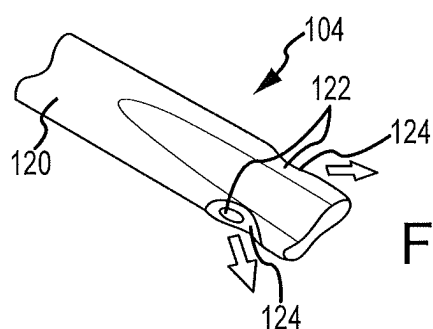
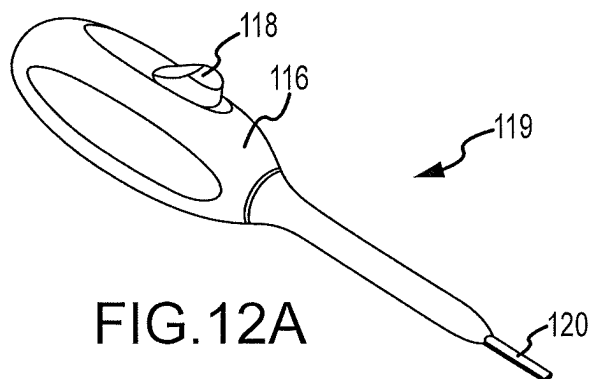


FIG. 12D

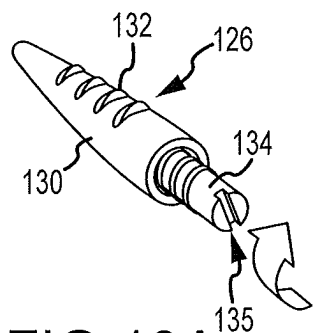


FIG. 13A

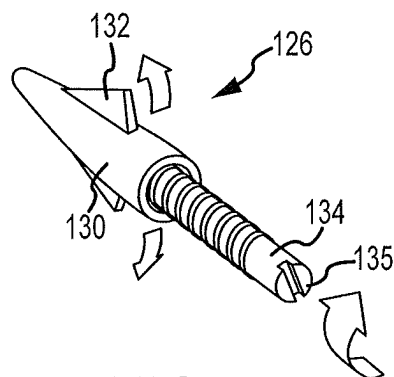


FIG. 13B

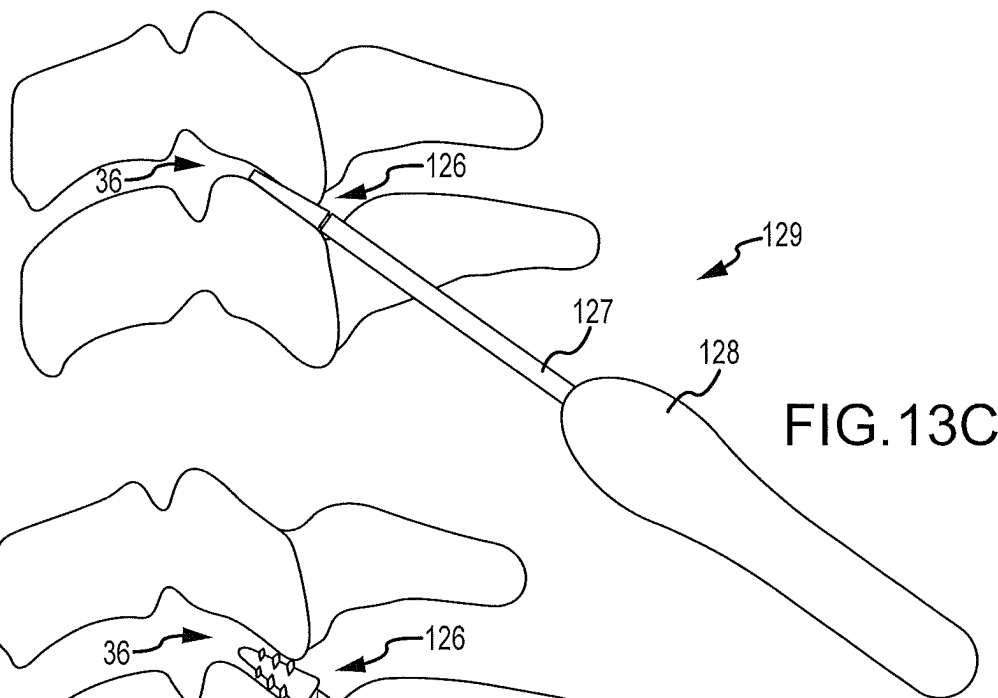


FIG. 13C

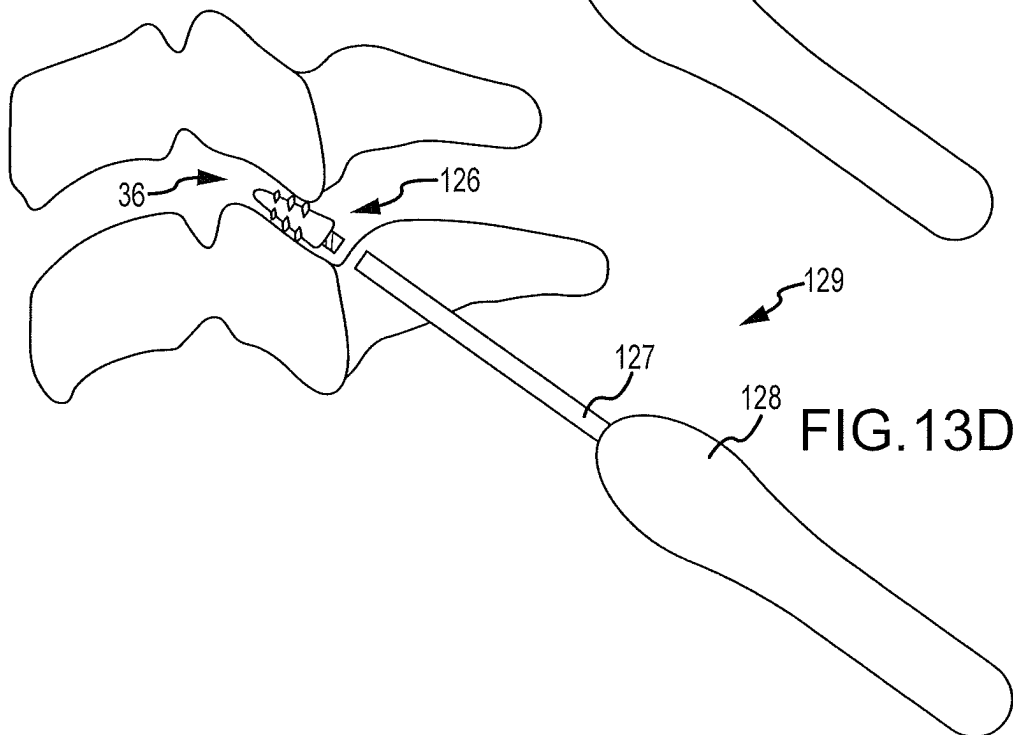


FIG. 13D

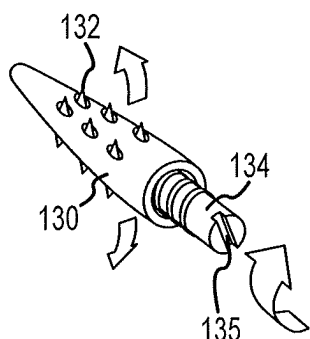


FIG. 13E

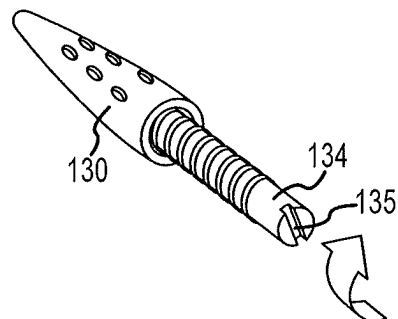


FIG. 13F

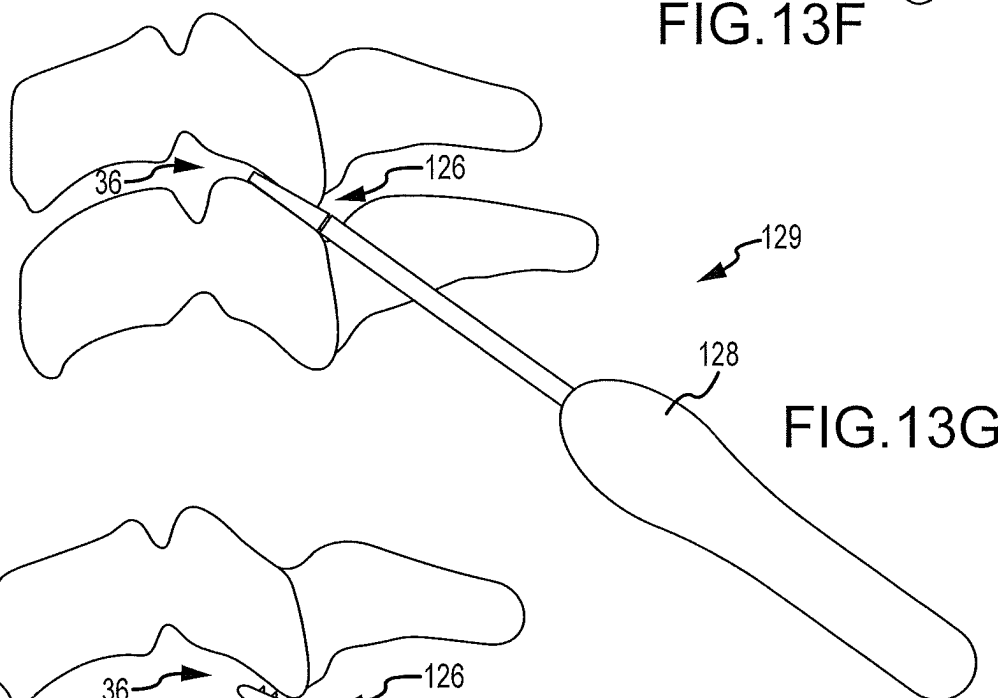


FIG. 13G

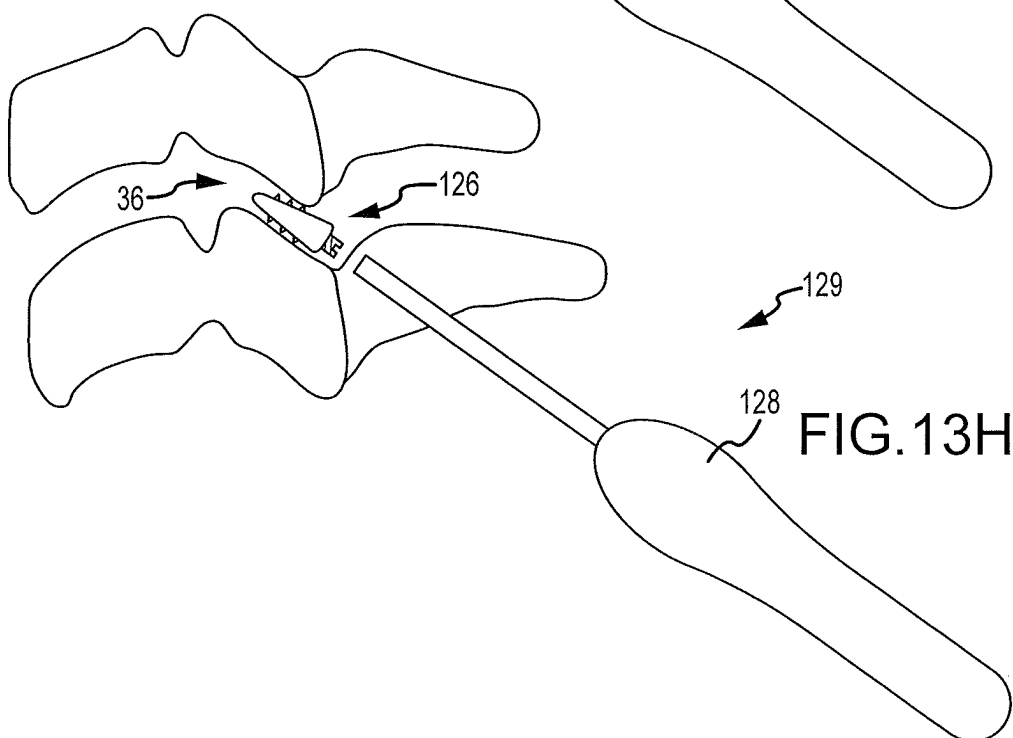
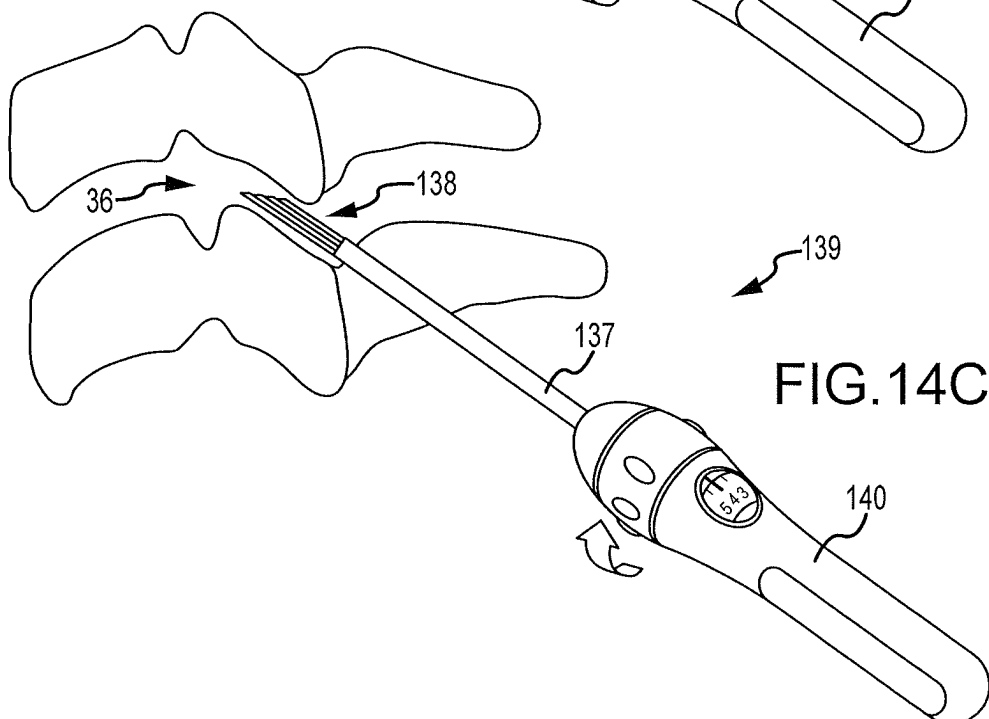
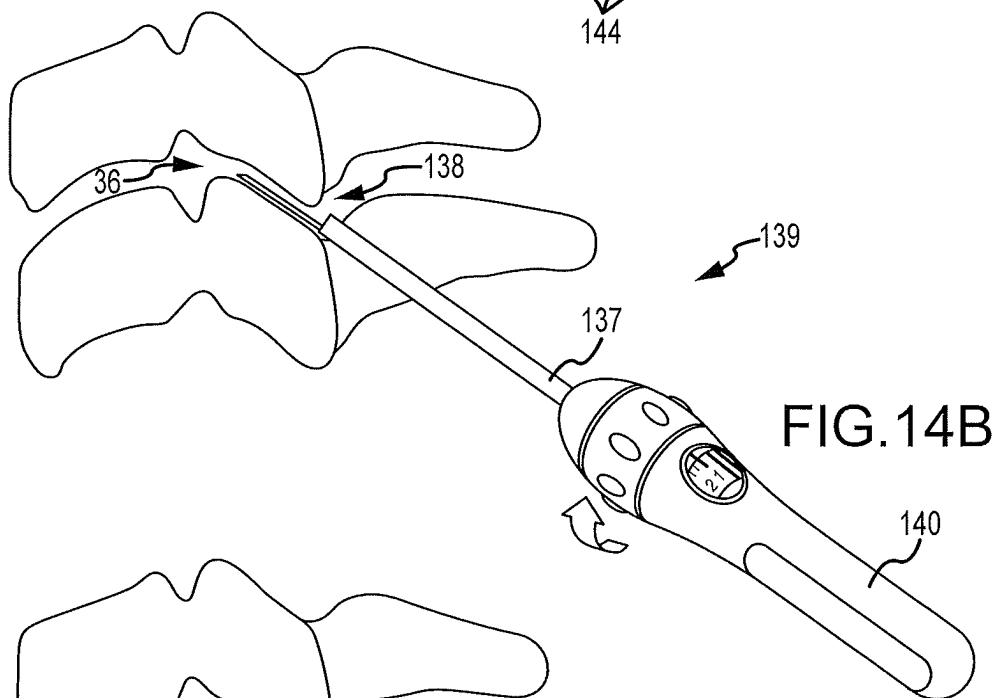
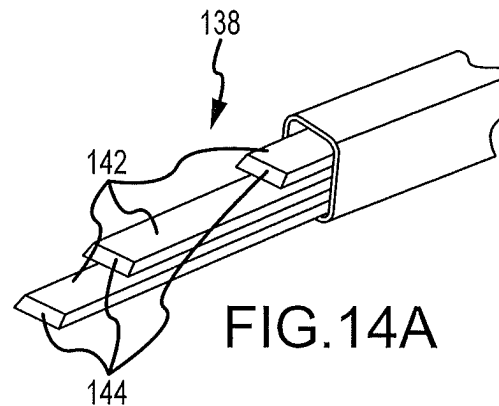


FIG. 13H



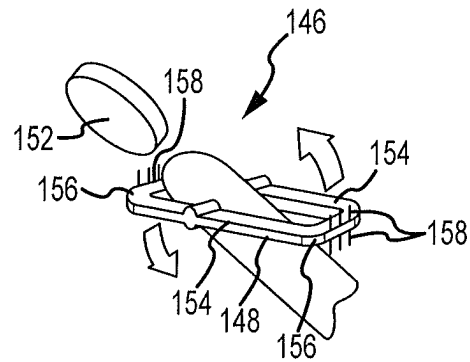


FIG. 15A

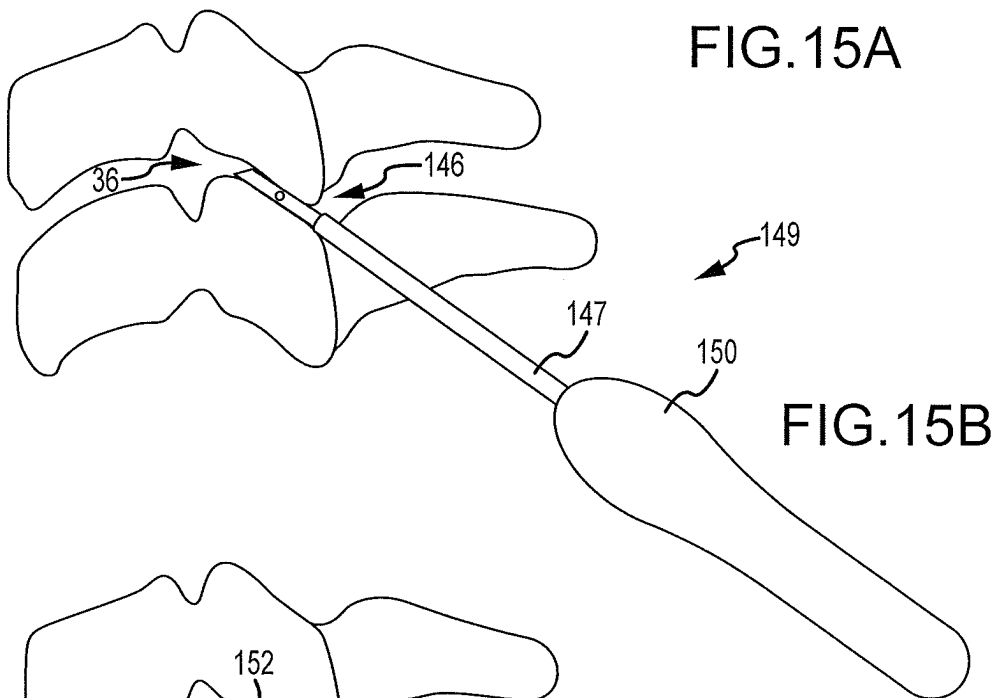


FIG. 15B

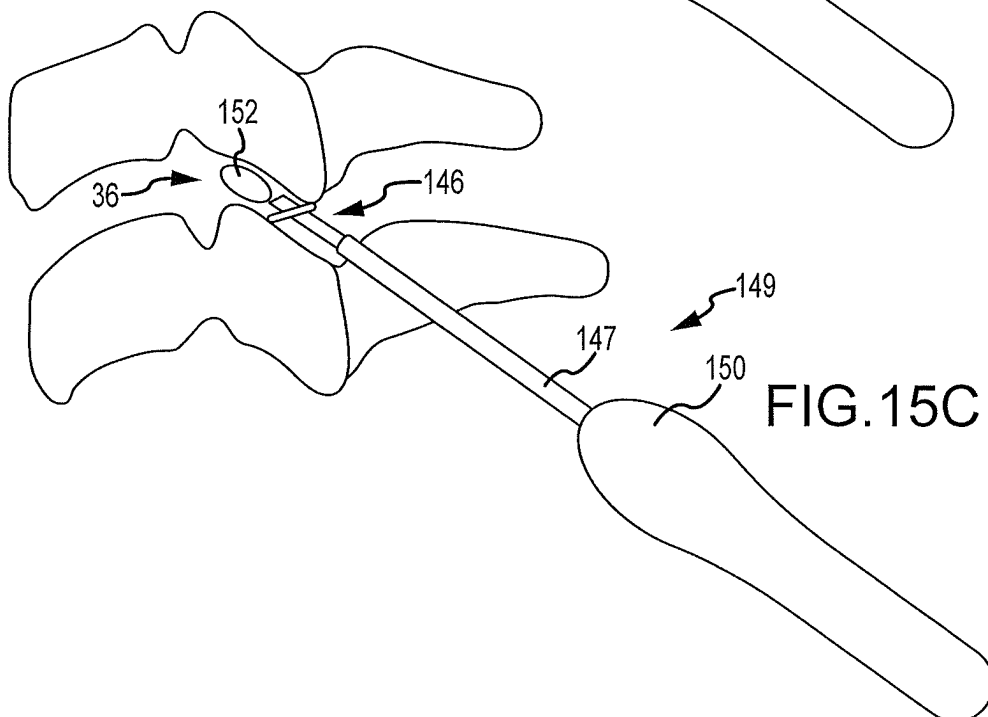


FIG. 15C

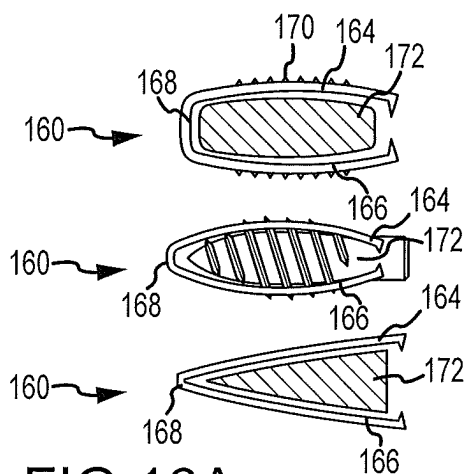


FIG. 16A

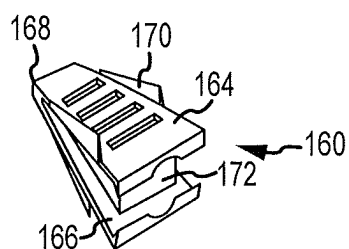


FIG. 16B

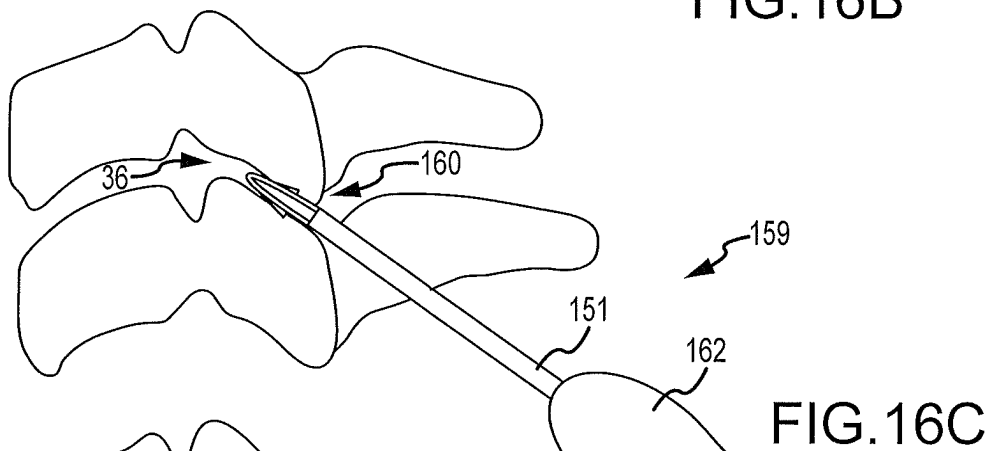


FIG. 16C

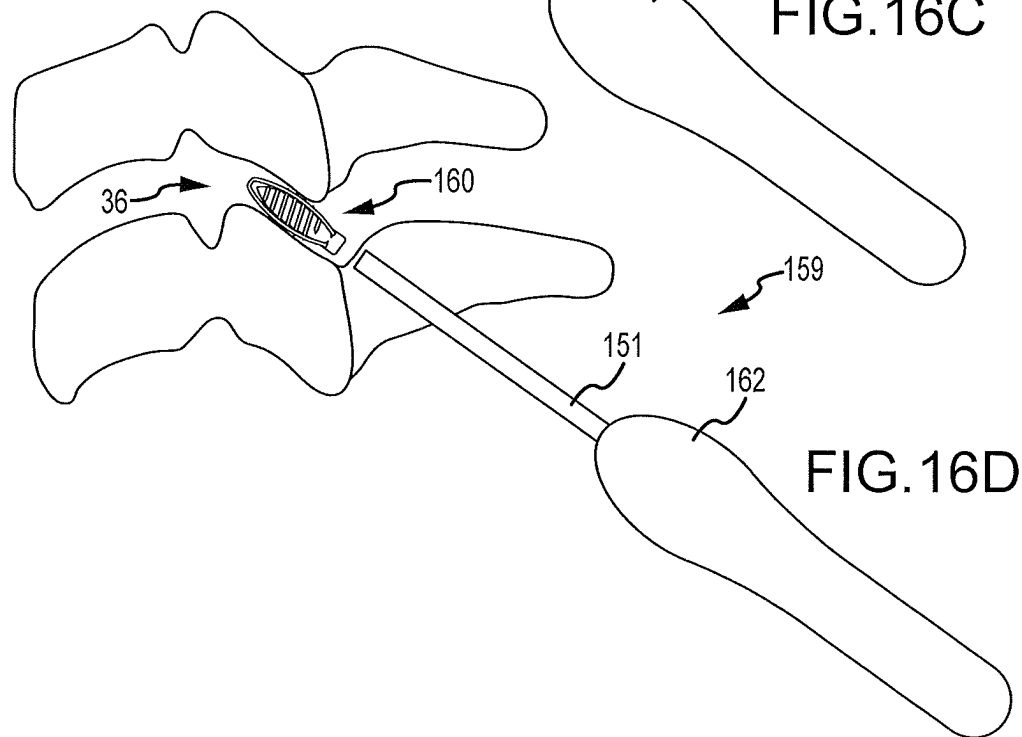


FIG. 16D

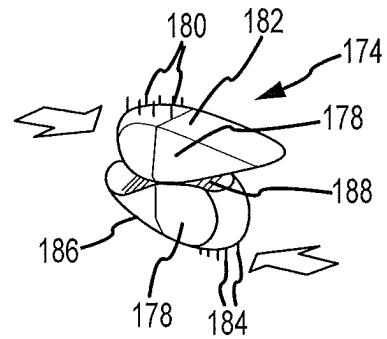


FIG.17A

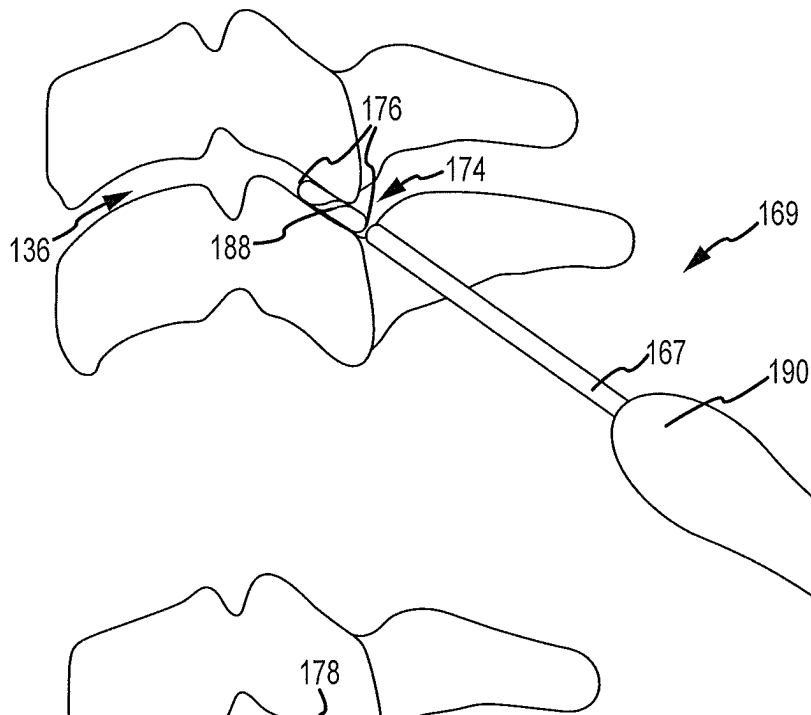


FIG.17B

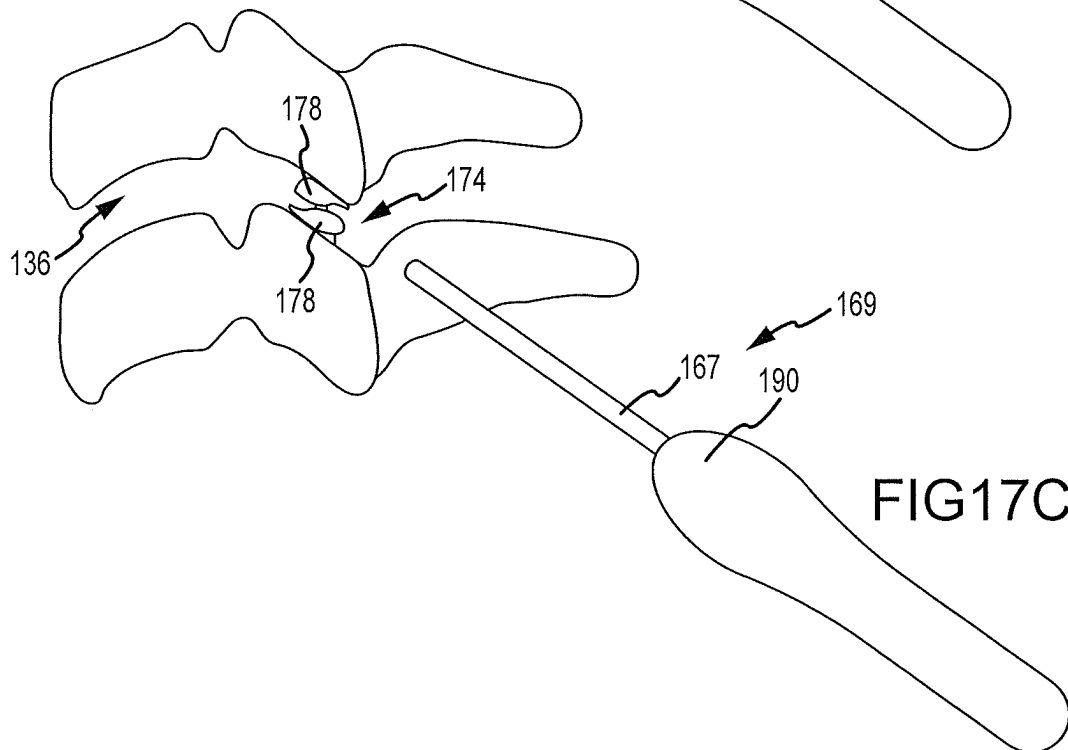


FIG.17C

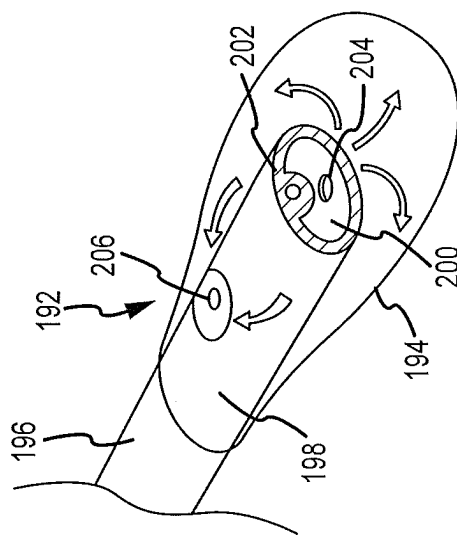


FIG. 18A

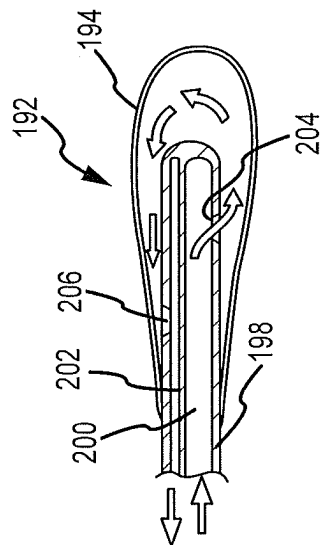


FIG. 18B

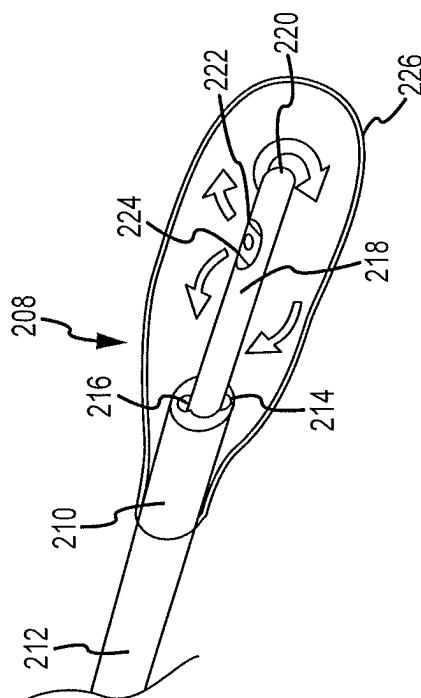


FIG. 19A

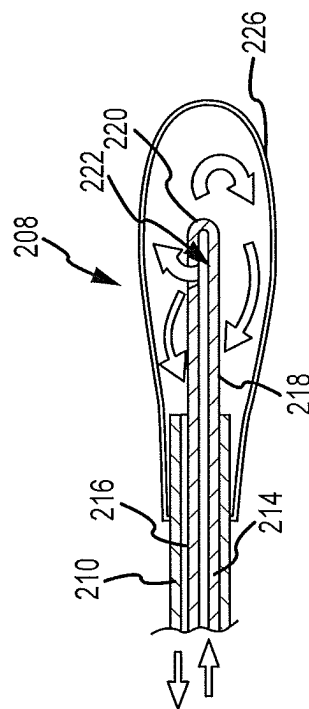


FIG. 19B

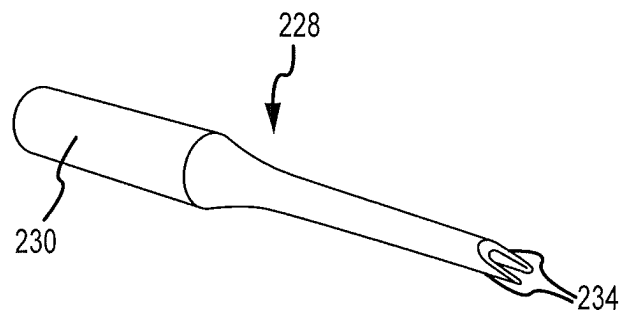


FIG. 20A

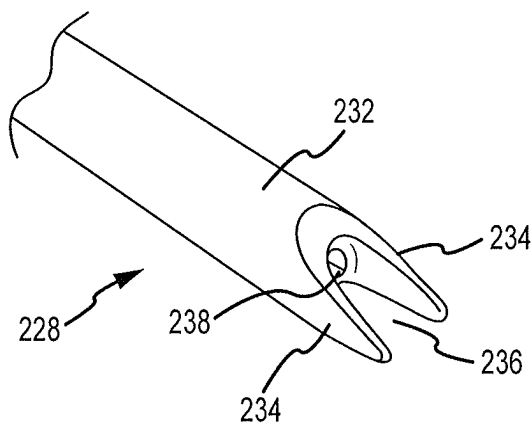


FIG. 20B

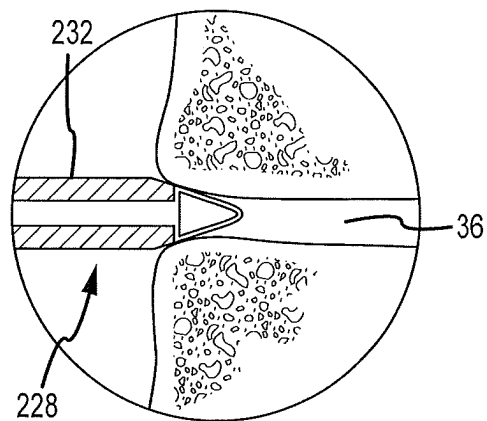


FIG. 20C

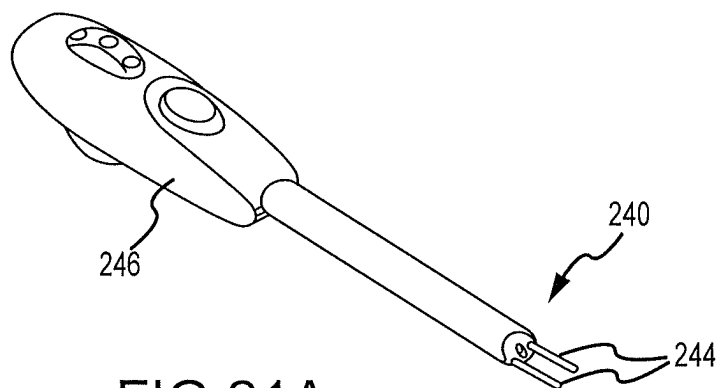


FIG. 21A

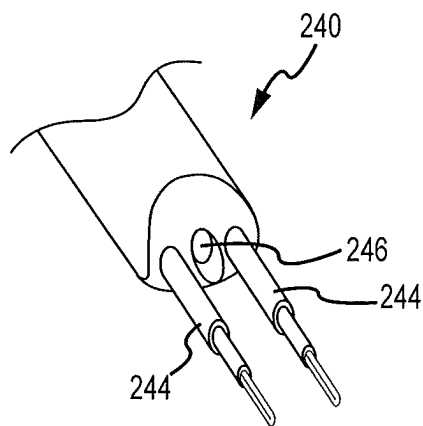


FIG. 21B

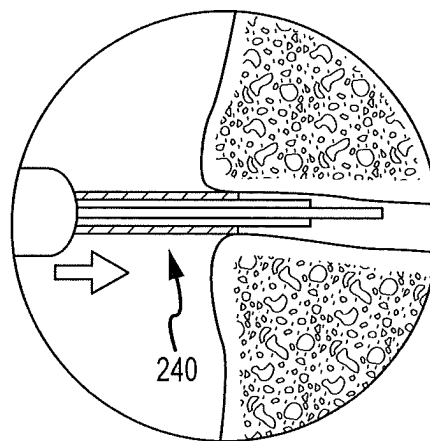


FIG. 21C

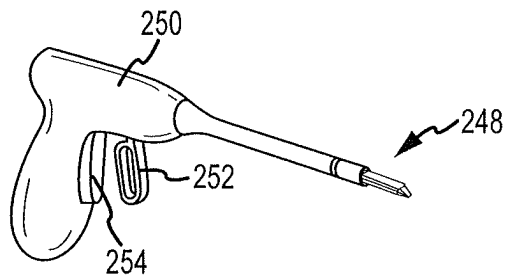


FIG. 22A

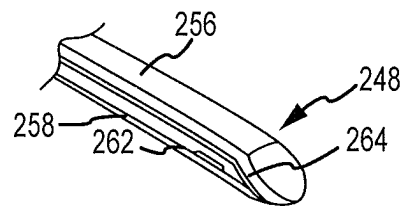


FIG. 22B

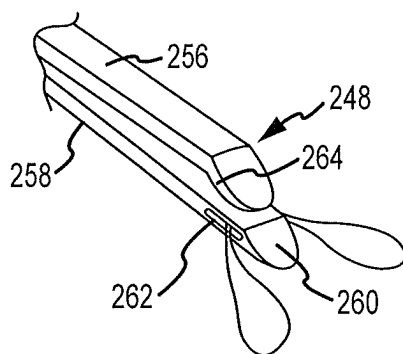


FIG. 22C

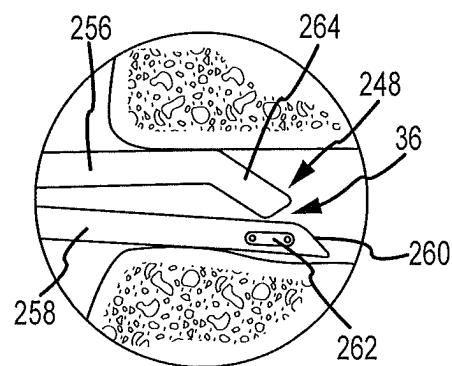


FIG. 22D

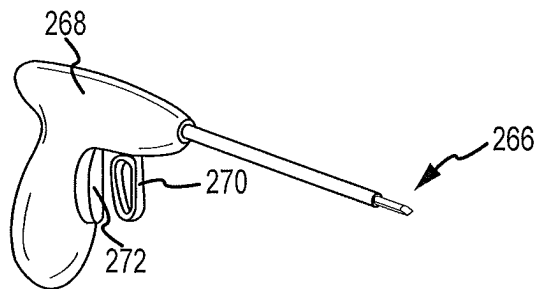


FIG. 23A

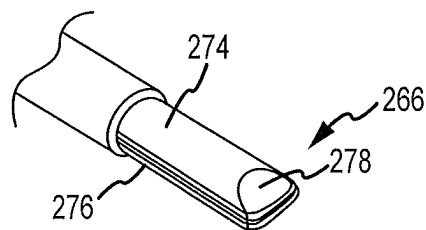


FIG. 23B

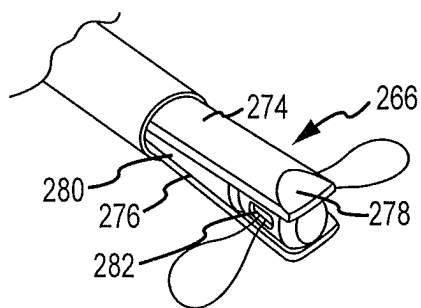


FIG. 23C

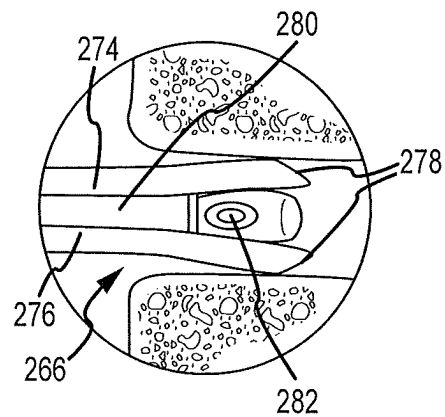


FIG. 23D

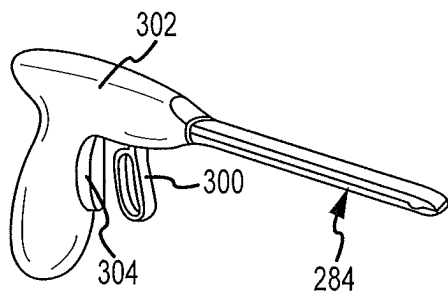


FIG. 24A

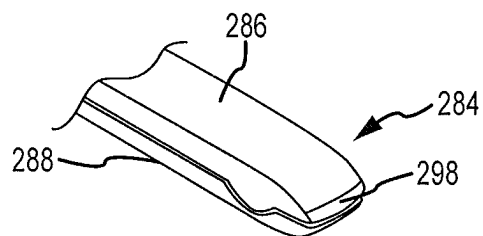


FIG. 24B

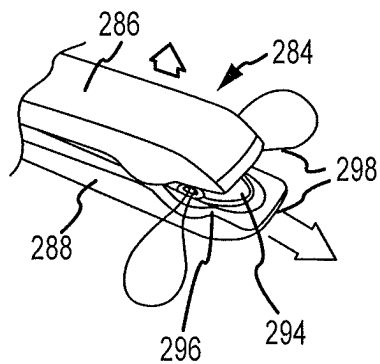


FIG. 24C

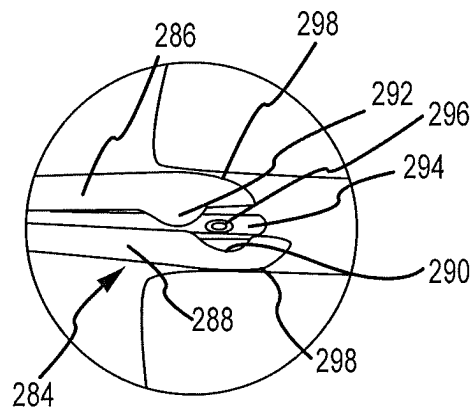


FIG. 24D

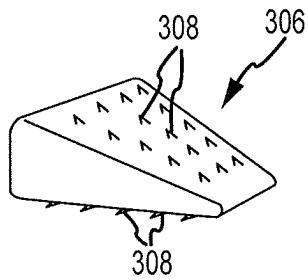


FIG. 25A

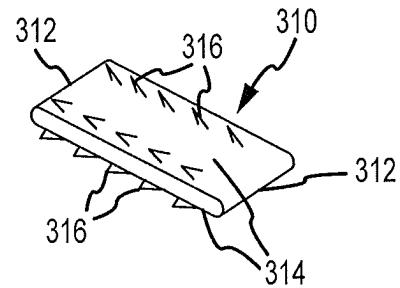


FIG. 25B

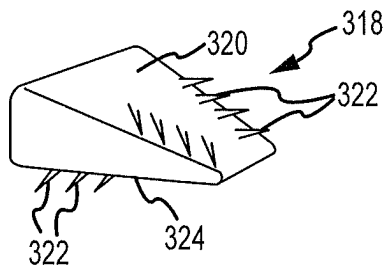


FIG. 25C

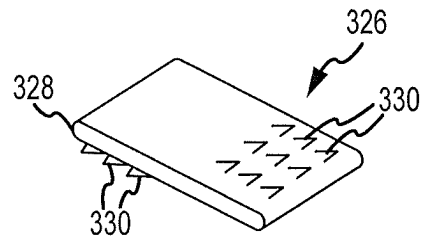


FIG. 25D

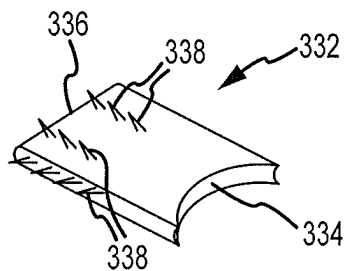


FIG. 25E

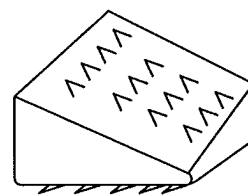
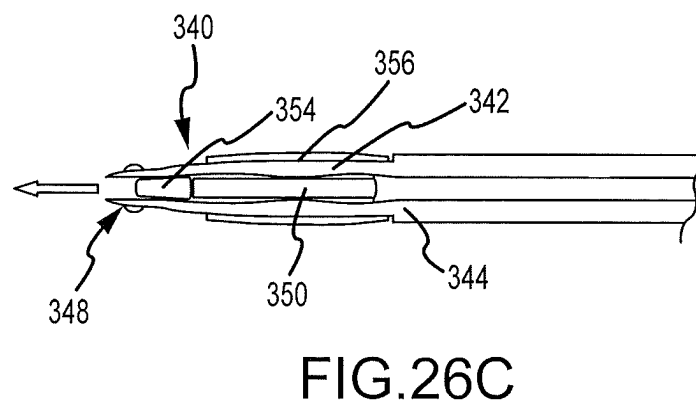
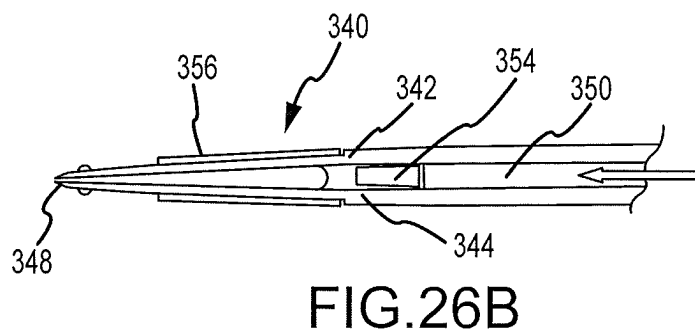
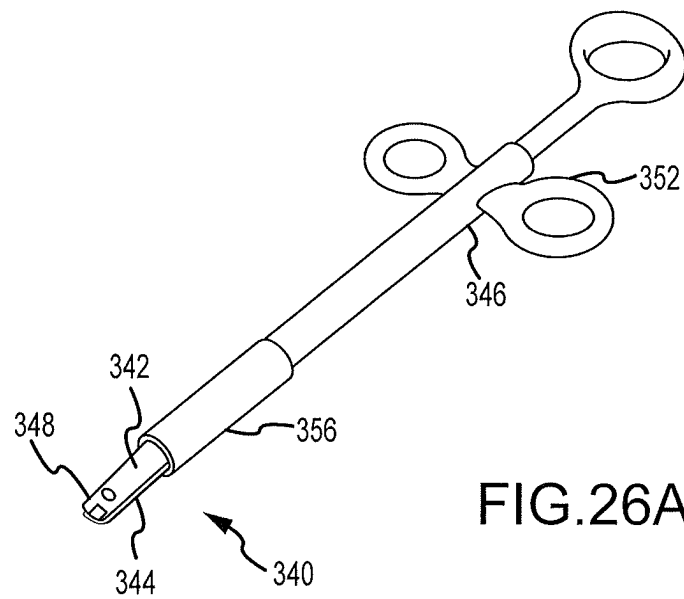


FIG. 25F



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CERVICAL DISTRACTION/IMPLANT DELIVERY DEVICE

CROSS REFERENCE TO RELATED APPLICATION

The present application is a continuation of U.S. Application No. 16,275,463 filed Feb. 14, 2019, now U.S. Pat. No. 11,224,521, which is a continuation of U.S. application Ser. No. 15/488,989 filed Apr. 17, 2017 and entitled Cervical Distraction/Implant Delivery Device (the '989 application), now U.S. Pat. No. 10,238,501. The '989 application is a continuation of U.S. application Ser. No. 13/949,042 filed Jul. 23, 2013, and entitled Cervical Distraction/Implant Delivery Device (the '042 application), now U.S. Pat. No. 9,622,874. The '042 application is a continuation of U.S. application Ser. No. 12/559,193 filed Sep. 14, 2009 and entitled Cervical Distraction/Implant Delivery Device (the '193 application), now U.S. Pat. No. 8,512,347. The '193 application is a continuation-in-part of U.S. application Ser. No. 12/455,814 filed Jun. 5, 2009, and entitled Facet Joint Implants and Delivery Tools (the '814 application), now U.S. Pat. No. 8,361,152. The '814 application is a continuation-in-part of U.S. application Ser. No. 12/317,682 filed Dec. 23, 2008, and entitled Facet Joint Implants and Delivery Tools (the '682 application), now U.S. Pat. No. 8,267,966.

The '193 application also claims priority from: U.S. Provisional Application No. 61/169,601 filed Apr. 15, 2009 and entitled Facet Joint Implants and Delivery Tools; U.S. Provisional Application No. 61/109,776 filed Oct. 30, 2008 and entitled Facet Joint implants; and U.S. Provisional Application No. 61/097,103 filed Sep. 15, 2008 and entitled Cervical Distraction/Implant Delivery Device.

The '682 application claims priority from U.S. Provisional Application No. 61/059,723 filed Jun. 6, 2008 and entitled Spine Distraction Device.

The contents of all of the above applications are hereby incorporated by reference herein in their entireties.

BACKGROUND OF THE INVENTION

Field of the Invention

This invention pertains generally to a cervical distraction device and more particularly to such a device that cannot only distract a facet joint but also deliver an implant to the distracted joint.

Description of the Relevant Art

Chronic back problems cause pain and disability for a large segment of the population. Adverse spinal conditions are characteristic of age. With aging, generally comes an increase in spinal stenosis (including, but not limited to, central canal and lateral stenosis), and facet arthropathy. Spinal stenosis results in a reduction of foraminal area (i.e. the available space for the passage of nerves and blood vessels), which compresses the cervical nerve roots and causes radicular pain. Extension and ipsilateral rotation of the neck further reduces the foraminal area and contributes to pain, nerve root compression, and neural injury. However, neck flexion generally increases the foraminal area.

Cervical disc herniations predominantly present upper extremity radicular symptoms. The vast majority of these herniations do not have an associated neurological deficit and present pain only. A well-described treatment for cer-

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vical disc herniations is closed traction. There are a number of marketed devices that alleviate pain by pulling on the head to increase foraminal height.

Cervical disc herniations have been treated with anterior and posterior surgery. The vast majority of these surgeries are performed through an anterior approach, which requires a spinal fusion. These surgeries are expensive and beget additional surgeries due to change in biomechanics of the neck. There is a three percent incidence of re-operation after cervical spine surgery.

It is an object of the present invention to provide a minimally invasive device and procedure to increase foraminal height to reduce radicular symptoms for patients with disc herniations.

SUMMARY OF THE INVENTION

In one embodiment, a distraction tool for distracting a facet joint can include a handle, a cannula extending from the handle, and a distraction mechanism positioned on a distal end of the cannula and adapted to be placed in the facet joint. In this embodiment, the handle can be in communication with the distraction mechanism and actuation of the handle can cause distraction thereof.

In another embodiment, a distraction mechanism for distracting a facet joint can be provided. The mechanism can include a head in the form of an oblong band and an elongate member extending through the head. The elongate member can be adapted to draw opposing ends of the head toward one another thereby expanding the head.

In still another embodiment, a distraction mechanism can include a plurality of pairs of teeth in pivotal relation with a central core and an elongate member extending through the central core. Actuation of the elongate member can cause distraction of at least one of the plurality of pairs of teeth.

In still another embodiment, a distraction mechanism can include a hollow elongated body having facet engaging features and an elongated member adapted for insertion within the body. Insertion of the elongated member can actuate the facet engaging features.

In still another embodiment, a distraction mechanism can include an expandable receiving portion having upper and lower generally planar elements connected at an end with a living hinge. The distraction mechanism can also include an actuation device adapted to be placed between the planar elements. The planar elements can include malleable material that conforms to the facet surface upon implantation.

In yet another embodiment, a delivery tool for positioning a distraction mechanism in a facet joint can include a distraction mechanism for distracting a facet joint and a handle and a cannula connected to one another and carrying said distraction mechanism on a distal end thereof. The handle can include means for delivering energy to said distraction mechanism, and said distraction mechanism can be manipulatable to increase the spacing of said confronting facets thereby distracting said facet joint.

A device and technique are disclosed for a minimally invasive surgical implantation to reduce radicular symptoms by inserting a distraction mechanism in a facet joint of an affected level of the spine to preserve the physiology of the spine. In particular, embodiments of the present invention provide for distracting and translating the cervical spine to increase the foraminal dimension in extension and neutral positions. The distraction mechanism may have a portion which can serve as an implant or It may be a mechanism for facilitating insertion of a separate implant. When the distraction mechanism is positioned in the cervical facet joint,

it expands to distract or increase the space between the vertebrae to increase the foraminal area or dimension and reduce pressure on the nerves and blood vessels of the cervical spine. The devices and techniques disclosed supplement those disclosed in U.S. nonprovisional patent application Ser. No. 11/618,619 filed Dec. 29, 2006, entitled Cervical Distraction Device, and U.S. provisional patent application Ser. No. 61/059,723, filed Jun. 6, 2008, entitled Spine Distraction Device, which are of common ownership with the present application, the disclosures of which are hereby incorporated by reference.

The implantation procedure may be performed under conscious sedation in order to obtain intra-operative patient symptom feedback.

After achieving the desired distraction of the facet joint, the distal tip of the distraction tool may be detached from the tool so the distraction device itself serves as a permanent implant for placement in the facet joint. The patient is left with the distraction device implant in the facet joint with permanent increased foraminal height. As an alternative, the distraction device can be removed from the distracted joint after a separate or auxiliary insert is positioned in the joint.

While the implant may comprise an inflatable balloon configured to be filled with an inflation medium, e.g. hydrogel or the like, to distribute a compressive load on the articulating surfaces as disclosed in the aforementioned U.S. nonprovisional patent application Ser. No. 11/618,619, pursuant to the present invention, the implant may also be a mechanical device that does or does not expand or inflate.

The implant is configured to dynamically stabilize or fuse the facet joint and retain it in an expanded or distracted condition. The implant maintains a minimal distance between the articulating surfaces and, in some embodiments, allows motion of a first vertebra with respect to a second adjacent vertebra.

According to the technique of the invention, an expandable or non-expandable distraction device is inserted in a collapsed state into a facet joint bounded by first and second vertebrae, and is expanded within the facet joint to increase a (foramina) dimension associated with the first and second vertebrae. The implant is installed in a facet joint located between adjacent cervical vertebrae. The expandable implant engages the articulating surfaces of the facet joint to increase the distance between the articulating surfaces.

Further aspects of the invention will be brought out in the following portions of the specification, wherein the detailed description is for the purpose of fully disclosing preferred embodiments of the invention without placing limitations thereon.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a fragmentary isometric of a portion of the human spine showing a facet joint which may be treated with the devices and techniques of the present invention to distract the facet joint and retain it in an expanded condition.

FIG. 2 is an isometric similar to FIG. 1 viewed from a different direction.

FIG. 3 is an isometric similar to FIG. 2 with a distraction device in accordance with the present invention having its distal end inserted into the facet joint and having an implant device releasably held in the distal end.

FIG. 4 is an isometric similar to FIG. 3 with the implant having been positioned within the facet joint to retain the distracted position of the joint.

FIG. 5 is a section taken through the spinal column showing implant in position on bilateral facets.

FIG. 6A is an isometric section with parts removed of a first embodiment of a distraction device in accordance with the present invention.

FIG. 6B is a diagrammatic side elevation of the device of FIG. 6A with the distal tip of the device positioned within a facet joint.

FIG. 6C is an elevation similar to FIG. 6B with the device expanded to distract the facet joint.

FIG. 7A is a fragmentary isometric of a second embodiment of the distraction device of the present invention.

FIG. 7B is a side elevation of the device of FIG. 7A having its distal tip of the device inserted into a facet joint.

FIG. 7C is an elevation similar to FIG. 7B with the distal tip expanded to distract the facet joint.

FIG. 8A is an isometric of an expandable tip of a distraction device showing a third embodiment of the present invention.

FIG. 8B is a side elevation showing the device of FIG. 8A positioned on the distal end of a distraction tool with the device inserted into a facet joint.

FIG. 8C is a side elevation similar to FIG. 8B with the device expanded to distract the facet joint.

FIG. 9A is a fragmentary isometric of the distal tip of a distraction device showing a fourth embodiment of the present invention.

FIG. 9B is a side elevation showing the device of FIG. 9A positioned in a facet joint.

FIG. 9C is a side elevation similar to FIG. 9B with the device expanded.

FIG. 10A is an isometric of a fifth embodiment of the device of the present invention.

FIG. 10B is an isometric similar to FIG. 10A with the device expanded.

FIG. 10C is a side view showing the device of FIG. 10A positioned on the distal end of a distraction tool.

FIG. 10D is a side view similar to FIG. 10C with the device expanded as in FIG. 10B and separated from the insertion tool.

FIG. 11A is a fragmentary isometric of a sixth embodiment of the present invention.

FIG. 11B is a side view showing the device of FIG. 11A inserted into a facet joint.

FIG. 11C is a side view similar to FIG. 11B with the device having been rotated with the insertion tool to distract the facet joint.

FIG. 12A is an isometric of the handle of a seventh embodiment of the present invention.

FIG. 12B is the distal tip of the seventh embodiment of the handle which is shown in FIG. 12A.

FIG. 12C is a transverse section of the distal tip as shown in FIG. 12B.

FIG. 12D is a diagrammatic cross-section through a facet joint with the device of FIG. 12B positioned therein and having injected a gel substance into the facet joint.

FIGS. 13A and 13B are fragmentary isometric views of a tip of a distraction device in accordance with the present invention showing an eighth embodiment of the distal tip of the device.

FIG. 13C is a side view of a tool inserting the device of FIG. 13A into a facet joint.

FIG. 13D is a side view similar to FIG. 13C with the device of FIG. 13A having been rotated to distract the facet joint and the insertion device removed from the tip which serves as an implant.

FIGS. 13E-13H are respectively the same views as depicted in FIGS. 13A-13D, except of a variation of the embodiment depicted in FIGS. 13A-13D.

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FIG. 14A is an isometric showing the tip of a ninth embodiment of a distraction device in accordance with the present invention.

FIG. 14B is a side view showing the distraction device of FIG. 14A inserted into a facet joint with an insertion tool.

FIG. 14C is a side view similar to FIG. 14B with a plurality of blades or spatula elements having been advanced into the facet joint to distract the joint.

FIG. 15A is an isometric of a tenth embodiment of a distraction device in accordance with the present invention.

FIG. 15B is a side view of the device shown in FIG. 15A on the distal end of an insertion tool and positioned within a facet joint.

FIG. 15C is a side view similar to FIG. 15B with the distraction device having been expanded to distract the facet joint and deliver a permanent implant into the distracted joint.

FIG. 16A is a side view of an eleventh embodiment of a distraction device in accordance with the present invention, showing a wedge option, an elongated member option, and a block option.

FIG. 16B is an isometric of the wedge option of FIG. 16A.

FIG. 16C is a side view of the distraction device of FIG. 16A positioned within a facet joint on the distal tip of an insertion tool.

FIG. 16D is a side view similar to FIG. 16B with the distraction device expanded to distract the facet joint and with the insertion tool having been separated therefrom.

FIG. 17A is an isometric of a twelfth embodiment of a distraction device in accordance with the present invention.

FIG. 17B is a side view of the device of FIG. 17A positioned in a facet joint and on the end of an insertion tool.

FIG. 17C is a side view similar to FIG. 17B with the device of FIG. 17A having been expanded and the insertion tool separated therefrom to leave the device as an implant in a distracted facet joint.

FIG. 18A is a diagrammatic fragmentary isometric with parts removed illustrating a thirteenth embodiment of the present invention.

FIG. 18B is a fragmentary diagrammatic vertical section through the device shown in FIG. 18A.

FIG. 19A is a fragmentary diagrammatic similar to FIG. 18A showing an alternative fourteenth embodiment.

FIG. 19B is a fragmentary vertical section similar to FIG. 18B showing the alternative embodiment of FIG. 19A.

FIG. 20A is an isometric of a fifteenth embodiment of a distraction device in accordance with the present invention.

FIG. 20B is an enlarged fragmentary isometric view of the distal tip of the device of FIG. 20A.

FIG. 20C is a diagrammatic vertical section showing the device of FIG. 20A inserted into a facet joint.

FIG. 21A is an isometric of a sixteenth embodiment of a distraction device in accordance with the present invention.

FIG. 21B is an enlarged fragmentary isometric of the distal tip of the device of FIG. 21A.

FIG. 21C is a diagrammatic vertical section showing the device of FIG. 21A inserted into a facet joint.

FIG. 22A is an isometric of a seventeenth embodiment of the distraction device of the present invention.

FIG. 22B is an enlarged fragmentary isometric of the distal tip of the device of FIG. 22A in a closed position.

FIG. 22C is a fragmentary isometric similar to FIG. 22B with the tip in an expanded position.

FIG. 22D is a diagrammatic vertical section showing the device in the expanded condition of FIG. 22C inserted into a facet joint.

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FIG. 23A is an isometric of an eighteenth embodiment of the distraction device in accordance with the present invention.

FIG. 23B is an enlarged fragmentary isometric showing the tip of the device of FIG. 23A in a closed position.

FIG. 23C is a fragmentary isometric similar to FIG. 23B with the tip in an expanded position.

FIG. 23D is a diagrammatic vertical section showing the device in the expanded position of FIG. 23C inserted into a facet joint.

FIG. 24A is an isometric of a nineteenth embodiment of the distraction device of the invention.

FIG. 24B is an enlarged fragmentary isometric of the tip of the device of FIG. 24A in a closed position.

FIG. 24C is a fragmentary isometric similar to FIG. 24B with the tip expanded.

FIG. 24D is a diagrammatic vertical section showing the device in the expanded position of FIG. 24C inserted into a facet joint.

FIG. 25A is an isometric of a first embodiment of an implant for delivery with or in conjunction with a distraction device.

FIG. 25B is an isometric of a second embodiment of an implant.

FIG. 25C is an isometric of a third embodiment of an implant.

FIG. 25D is an isometric of a fourth embodiment of an implant.

FIG. 25E is an isometric of a fifth embodiment of an implant.

FIG. 25F is an isometric of a sixth embodiment of an implant.

FIG. 26A is an isometric of a twentieth embodiment of the distraction device in accordance with the present invention.

FIG. 26B is a vertical section through the distal tip of the device of FIG. 26A in a closed position.

FIG. 26C is a vertical section similar to FIG. 26A with the distal tip expanded.

DETAILED DESCRIPTION OF THE INVENTION

Pursuant to the present invention, devices and techniques for distracting and retaining a facet joint in a distracted and forwardly translated condition are disclosed. Prior to distracting the facet joint, the joint, which can be difficult to access, can be accessed pursuant, for example, to the method and apparatus disclosed in U.S. Non-provisional application Ser. No. 12/350,609, filed Jan. 8, 2009, which is commonly owned with the present application and hereby incorporated by reference. Pursuant to the disclosure in that application, the access system is comprised of one or more cannulas made of steel, titanium, or plastic. The initial facet joint access cannula can have a sharp spatula tip on the distal end. The spatula tip can have a flat configuration to enable access into the flat facet joint. Once the spatula tip achieves access into the generally flatly oriented facet joint, subsequent stylets and working instruments can be passed down this access channel to complete a distraction procedure. The distraction procedure can be accomplished with devices and techniques to be described hereafter.

The percutaneous distraction mechanism can be introduced down the working cannula of the above-identified access system. The mechanism can be part of a delivery tool that would allow the surgeon to generate distraction by applying energy to a handle of the delivery tool for the distraction mechanism positioned at the distal end of the

tool. The handle of the delivery tool can be configured in any number of ways including but not limited to the following:

a) Trigger grip—index finger activates distraction by pulling the trigger to apply energy to the distraction mechanism.

b) Scissor grip—index and middle fingers meet and separate to apply energy to the distraction mechanism.

c) Thumb wheel or slide—thumb rolls a wheel or slides a slide that progressively applies more energy to the distraction mechanism.

d) Thumb cushion rod—thumb plunges a stylet down the working cannula to apply energy to the distraction mechanism.

e) Stylet screwdriver—stylet is threaded down the working cannula into the distraction mechanism applying increasingly more energy to the distraction mechanism as the stylet screwdriver advances.

f) Mallet based handle—a stylet with a flat malleable surface is inserted for the purposes of receiving and dispersing mallet energy and applying it to the distraction mechanism.

g) Thumb button—a button on the proximal end of the handle is pushed which creates one of a number of mechanical systems to apply energy to the distraction mechanism. Those mechanical systems could include but are not limited to:

- i) hydraulic pressure generation;
- ii) mechanical drill;
- iii) level system; or
- iv) elastic bands with “rope and pulley” mechanism.

h) Wedge firestarter—triangular wedge located at the proximal end of the tool is flattened to generate energy to apply to the distraction mechanism.

i) Foot or hand pump—feet or hands of surgeon used to press the system to create energy to be applied to the distraction mechanism.

Referring to FIGS. 1-5, a description relating to any and/or all of the delivery tools and associated distraction mechanisms and implants disclosed herein is presented. Referring first to FIG. 1, a portion of a spinal column 30 is shown having facets 32 on vertebrae 34 and with facet joints 36 between adjacent facets 32 of the vertebrae 34. The spinal cord 38, of course, passes vertically through the aligned vertebrae 34 with peripheral nerves 40 passing from the spinal cord 38 outwardly through the spinal column 30 through foraminal openings 42 to predestined locations in the human body. When facet joints 36 become narrowed, usually from disc degeneration, the foraminal openings 42 are reduced in size pinching the nerve and causing pain to the individual.

Pursuant to the present invention, the facet joint 36, for example as shown in FIG. 2, can be accessed using a system, for example, of the type described in the aforementioned U.S. Non-provisional patent application Ser. Nos. 11/618,619 and 12/350,609, and after gaining access to the facet joint 36, a delivery tool 46 shown by way of example in FIG. 3, can have its distal end inserted into the facet joint 36 and by expanding a distraction mechanism at the distal tip of the tool 46, the facet joint 36 can be distracted or enlarged. The distraction mechanism itself can be detachable from the tool 46 and left in the facet joint 36 as an implant or a separate implant can be inserted with the delivery tool 46 or otherwise once the joint 36 has been distracted. Accordingly, it is noted that while the tools herein are referred to as delivery tools because they may be used to deliver a distraction mechanism, the distraction mechanism may or may not be the resulting implant. FIG. 4 shows from a lateral location an

implant 48 positioned in a facet joint 36 which holds the joint 36 in a distracted condition, while FIG. 5 is a plan view showing implants in position on facets 32 of a vertebra 34.

A first embodiment of a delivery tool 59 in accordance with the present invention is shown in FIGS. 6A-6C. It will there be seen, pursuant to the above description with any and/or all of the embodiments of the present invention to be disclosed hereafter, a delivery tool can be inserted down the working cannula, for example of the access system described previously, which can be docked in a facet joint 36. The distal end of the delivery tool can be positioned at the anterior aspect of the joint 36 and the surgeon can apply energy to the delivery tool to create separation and distraction of the facet joint 36. The separation can occur in both the vertical and horizontal planes of the joint 36 resulting in vertical distraction and forward/anterior translation of the superior vertebrae relative to the inferior vertebrae. The facet joint distraction and forward translation can cause an increase in foramina) area and thus reduce nerve root compression and associated symptoms.

Referring to the first embodiment of the invention shown in FIGS. 6A-6C, a delivery tool 59 can include a handle 57, a distraction cannula 52, and a distraction mechanism 50. The handle portion 57 can include any of the energy delivering handles described above. The handle 57 can be connected to the cannula 52 and the distraction mechanism 50 positioned within the cannula 52. The handle 57 can be configured to actuate the distraction mechanism 50.

The distraction mechanism 50 can be positioned at the tip of the delivery tool 59 and can include a band 54 of a somewhat rigid material that can be flexed so as to expand and become tall and generally flattened as it exits a rectangular opening 56 in the wall 58 of the distraction mechanism 50. The proximal end of the band 54 can be moveable and the distal end can be restrained. The rectangular opening 56, and thus the corresponding protruding band 54 can be smaller in length and width than the dimensions of the facet joint 36 being treated. The band 54 can be made of titanium, carbon, PEEK, nitinol, allograft, polymer, or plastic. Other elastic materials can be used.

In use, the handle 57 can be actuated thereby advancing the band 54. The force on the band 54 together with the restrained distal end of the band 54 can cause the band 54 to change its shape and configuration. That is, the compression developed in the band 54 can cause it to buckle out of plane and bow upwardly through the opening 56 and allow it to flatten out or conform to surface profile of the articular surface of the superior facet. While the band 54 may be sufficiently flexible to bow under compression, it may be sufficiently rigid to apply energy or force to the contacted articular surface. This applied force can result in the distraction and forward translation of the joint 36.

In one embodiment, the distraction mechanism 50, once expanded, can be separated from the cannula 52 and remain in place as an implant to retain the expanded condition of the facet joint 36. This may occur by way of a circumferential perforation between the cannula 52 and the distraction mechanism 50 just proximal to the opening 56. A release latch can also be provided between the cannula 52 and the distraction mechanism 50. Accordingly, the cannula 52 can be released from the distraction mechanism 50 and handle 57 and cannula 52 can be removed leaving the distraction mechanism 50 behind. In an alternative embodiment, a separate implant of the types to be described hereafter in other embodiments of the invention could be positioned in the distracted facet joint 36 before the distraction device was removed.

A second embodiment of the invention is shown in FIGS. 7A-7C. A handle 67 and cannula 62 can be provided as described previously. In this embodiment, a distraction mechanism 60 in the form of a collapsible box can be positioned on the distal end of the delivery tool 69. The distraction mechanism 60 can include upper 66 and lower 64 walls as well as end walls 68 with the walls 64, 66, 68 being pivotally connected so the device is movable between a flattened position as shown in FIG. 76 to an expanded position as shown in FIG. 7A. A locking mechanism can also be provided such as a cross-brace or other device for maintaining the mechanism 60 in an expanded position once expanded.

Fixation mechanisms 70 can be provided on the exterior surface of both upper 64 and lower 66 walls. These fixation mechanisms 70 can be in the form of 1) Aggressive shark teeth, 2) Cleats, and/or 3) Roughened pores. The aggressive shark teeth (as shown in FIG. 7A) can have a directional orientation positioned to achieve optimal fixation relative to the natural biomechanics of various sections of the spine. The teeth can be long enough to gain purchase in the cortical bone of the facet surfaces. The cleats can have a less aggressive profile than the shark teeth but still allow for directional orientation for the same reasons described above. These cleats can also be capable of anchoring in the hard cortical bone of the facet surface. The roughened pores be positioned on the surface and can be adapted to prevent free sliding of the facet joint 36. These surfaces can be roughened and coated with FDA master file approved resurfacing chemicals that create friction and prevent device migration.

The collapsible box can be made of titanium, steel, carbon, PEEK, nitinol, polymer, or plastic. As with the embodiment described above, the collapsible box system can be configured to detach for permanent implantation or can be used to retain distraction of the joint 36 while an auxiliary implant or gel is positioned in the distracted joint 36.

In use, the collapsible box can be inserted through the cannula of an access system as described previously to position the collapsible box within the facet joint 36. The box can be actuated with the handle 67 of the tool causing the box to transition from a collapsed or flat parallelogram configuration to an expanded rectangular configuration. As is probably best appreciated by reference to FIG. 7A, the collapsible box can be moved between its collapsed and expanded conditions by actuating the handle 67, which can pull or push on an upper flexible but somewhat rigid strip 72 connected to the upper wall 64 while a lower similar strip 74 connected to the lower wall 66 holds the lower wall 66 in a fixed position. The expansion can create separation of the facet joint 36 in both the vertical and horizontal planes. As the collapsible box expands and causes distraction and translation, the fixation mechanisms 70 engage the facet surfaces securing the structure and allowing for controlled distraction and translation of the facet joint 36.

A third embodiment of a delivery tool 79 in accordance with the invention is shown in FIGS. 8A-8C and can be seen to be similar to that of the second embodiment of FIGS. 7A-7C. In this embodiment, a distraction mechanism 76 in the form of a collapsible box is again utilized but the bottom surface of the lower wall 78 of the box has fixation mechanisms 80 for engaging the lower facet to hold the lower wall 78 of the box in a fixed position while the upper wall of the box with no such teeth is shifted relative thereto with a pair of rigid but somewhat flexible strips 84 that can be extended or retracted with the insertion handle 77 to expand or flatten the collapsible box, respectively.

A fourth embodiment of a delivery tool 89 in accordance with the present invention is shown in FIGS. 9A-9C. In this embodiment of the invention, the distraction mechanism 88 can have a diamond-shaped or oblong distraction head 90 made from a somewhat rigid but flexible band or strip of material. The mechanism 88 can include an actuator in the form of an elongate member 92 extending longitudinally there through that is adapted to draw the opposing ends 94, 96 of the distraction head 90 together. The elongate member 92 can be a threaded member adapted to draw the ends 94, 96 of the head 90 together by interaction with a female member 91. The interaction can be via a screw action with a female threaded member as shown. Alternatively, the elongate member 92 can be a toothed member adapted to draw the ends 94, 96 of the head 90 together via a ratcheting action with a female ratchet member similar to that of a cable tie. In either case, the elongate member 92 can extend through the head 90 of the distraction mechanism 88 and can include a stop or flange 93 on a distal end preventing the head 90 from advancing beyond the distal end of the elongate member 92. The female member 91 can be positioned on the elongate member 92 just proximal to the head 90. Advancing the female member 91 along the elongate member 92 via screwing action or sliding ratcheting action can compress the head 90 between the female member 91 and the flange 93 thereby causing the head 90 to expand as shown when comparing FIG. 9B to 9C.

It is noted that several alternative configurations and relationships of the elongate member 92 with a female member 91 can be provided. That is, for example, one end of the head 90 can form the female member 91 such that actuation of the elongate member 92 causes the end of the head 90 with the female member 91 to wait along the elongate member 92 thereby drawing the ends 94, 96 of the head together. Other configurations and relationships can be provided and are within the scope of the present invention.

It is also noted that FIGS. 9A-9C depict a female member 91 in the form of a nut that is threadable over an elongate member 92 in the form of a bolt. Accordingly, the cannula 86 of the delivery tool 89 includes a distal tip in the form of a nut driver. Those of skill in the art will understand and appreciate that alternative engagements between the cannula 86 and the female member 91 or the elongate member 92 can be provided. That is, depending on the configuration, actuation may require that the female member 91 be rotated or advanced and in other configurations, the elongate member 92 may need to be rotated, advanced, or even withdrawn. In some embodiments, the engagement between the cannula 86 and the female member 91 or elongate member 92 can be, for example, a straight screw driver tip, an alien wrench type tip, or other engaging shapes.

The head 90 of the distraction mechanism 88 can be made out of titanium, steel, carbon, PEEK, nitinol, or plastic. Other materials can be used. The distraction mechanism 88 can be configured to detach from the delivery tool 89 as shown so that it becomes a permanent implant. Alternatively, the distraction mechanism 88 can be used only for distraction purposes so that a separate implant can be positioned in the facet joint 36 to retain the distraction while the distraction mechanism 88 is removed. In this embodiment, the cannula 86 can include a more permanent connection to the female member 91 or elongate member 92 in contrast to that depicted in FIGS. 7B and 7C.

In use, energy can be applied to the flattened diamond or oblong head 90 via the handle 87 and cannula 86 assembly. The energy can cause the opposing ends 94, 96 of the head 90 to draw together and the head 90 can expand resulting in

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an increased height in the head **90**. The expansion of the head **90** can cause the surfaces of the head **90** to engage with the facet surfaces. As the head **90** expands against the facet surfaces, the joint **36** can separate in both the horizontal and vertical planes. This separation can lead to distraction and forward translation of the facet joint **36**. Moreover, the flexible nature of the head **90**, while sufficiently rigid to cause separation of the joint **36**, may also conform to the contour of the articular surfaces of the facet joint **36** thereby distributing the compressive load from the joint **36** more evenly over the surface interacting with the distraction mechanism **88**. This shape conformance can also function to resist withdrawal or dislodgement of the implant. Additionally, while not shown, the head **90** of the device can include fixation mechanisms along the surface of the head **90** adapted to engage the articular surfaces of the facet joint **36**.

A fifth embodiment of the distraction mechanism of the present invention is shown in FIGS. **10A-10D**. In this embodiment, pairs of pivotal teeth **100** can be mounted on a common base or central core **95**. The teeth **100** can be pivotal between a folded or retracted position as shown in FIG. **10C** or the teeth **100** can be pivoted outwardly. Each pair of teeth **100** may be pivotal independent of the other teeth **100**. However, as in the embodiment shown, the teeth **100** may overlap in the retracted position such that when underlying teeth **100** are pivoted outwardly the overlying teeth **100** are naturally lifted. The retracted position can provide for insertion of the mechanism **98** into a facet joint **36** and the outwardly pivoted position can serve to distract the joint **36**.

The distraction mechanism **98** can also include an actuation device in the form of an elongate member **101** adapted to be advanced into the base or central core, where advancing the elongate member **101** can serve to actuate the teeth **100** and pivot them outwardly. As shown, the elongate member **101** can be a threaded member and the central core **95** can include thread slots positioned below the position of the retracted teeth **100** and relatively close to the pivot point of the teeth **100**. Accordingly, when threaded member is advanced into central core **95**, the threads from the threaded member can protrude through the thread slots and can engage the teeth **100** relatively close to the pivot point of the teeth **100**. The protruding thread can thus distract the teeth **100** and pivot them outward. As the threaded member is continually advanced, additional teeth **100** can be distracted thereby distracting the facet joint **36**. The close proximity of the protruding thread to the pivot point of the teeth **100** can function to minimize the distance that the thread must protrude to suitably distract the associated teeth **100**.

In an alternative embodiment, the elongate member **101** may be a longitudinal shaft with radiused fins positioned along two lateral sides of the shaft. Each pair of fins can be positioned to correspond to each pair of teeth **100** and the fins can be positioned longitudinally along the elongate member **101** so as to be in close proximity to the pivot point of a respective pair of teeth **100**. The elongate member **101** can be rotated causing the fins to pass beneath the teeth **100** near their respective pivot points and cause the teeth **100** to simultaneously pivot outward.

In still another embodiment, the elongate member **101** can be in the form of plunger type actuation device. In this embodiment, the teeth **100** may extend through the base or central core **95** via a slit in the base or central core **95**. The teeth **100** can further extend to and be pivotally connected to the elongate member **101** passing through the central core **95**. Accordingly, advancing the elongate member **101** forward or distally can cause the teeth **100** to retract due to

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decreasing the angle of the pivot point of the elongate member **101** relative to the slit through which it passes. Withdrawing the elongate member **101**, on the other hand, can cause the teeth **100** to pivot outwardly simultaneously thus functioning to distract the joint **36**. In this embodiment, the elongate member **101** can threadably engage the inside of the central core **95** to allow for controlled advancement and withdrawal of the elongate member **101** or the elongate member **101** can be slidably received in the central core **95**.

Any or all of the above described elongate members **101** can include an engagement feature at a proximal end for engagement by the cannula **105** or device positioned within the cannula **105**. The engagement feature can be adapted to provide for transferring rotational, advancing, or withdrawing forces. As shown, the engagement feature can include a straight screw driver receiving slot **103**. The engagement feature can be an alien type connection or a hex head for receiving a nut driving device, or a square head for receiving a square drive device. Those of skill in the art will understand and appreciate that several engagement features are available and are within the scope of the invention.

When energy is applied to the handle **97** of the tool **99**, the teeth **100** of each pair can change their orientation or expand to achieve a suitable angle relative to the central core **95**. In some embodiments this angle can range from approximately 10° to approximately 90°. In other embodiments, this angle can be approximately 45°. In this embodiment, in its fully expanded state, the mechanism **98** can take on the appearance of a Christmas tree.

In use, the mechanism **98** can be inserted into the facet joint **36** in a flattened, collapsed state. Distraction energy can be applied to the handle **97** causing the mechanism **98** to expand which causes the multiple teeth **100** to engage both the top and bottom facet surfaces of the joint **36**. As the mechanism **98** is expanded to achieve increasingly larger dimensions, the facet joint **36** surfaces can separate. This separation in both the vertical and horizontal planes of the facet joint **36** can cause distraction and translation of the facet joint **36**.

A sixth embodiment of a tool **109** of the present invention is shown in FIGS. **11A-11C**. In this embodiment, the distraction mechanism **102** at the distal end of the tool **109** can have a relatively flat and ovular head **110** with a beveled tip **106** to facilitate insertion into the facet joint **36**. The mechanism **102** can have opposing relatively flat surfaces where the edges **112** interconnecting the surfaces are rounded along the lateral and medial edges of the mechanism **102**. The distraction mechanism **102** can be connected to the distal end of the cannula **107** with a rotationally resistive connection such that rotation of the cannula **107** causes rotation of the mechanism **102**.

In use, distraction energy can be applied to the handle **108** of the delivery tool **109** to cause rotation of the distraction mechanism **102**. As the flat head **110** of the mechanism **102** rotates, the articular surfaces of the facets can be forced apart due to a height of the mechanism (through the width of the head **110**) being greater than its flattened dimension. The increased height achieved from rotation of the mechanism **102** can cause the flat, round surfaces of the mechanism **102** to engage the facet surfaces and separate them. This rotational distraction can result in vertical and horizontal separation of the facet joints **36** achieving distraction and forward translation of the joint **36**.

The distraction mechanism **102** can be mounted on the distal end of the delivery tool **109** so that it can be removed from the delivery tool **109** if desired to remain as an implant to retain the distraction of the joint **36** or can be retained in

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position until an auxiliary implant is positioned in the joint 36 and then removed with the delivery tool 109.

A seventh embodiment of the present invention is shown in FIGS. 12A-12D. This embodiment is similar to that of the sixth embodiment in that the mechanism 104 is rotatable to achieve distraction of the joint 36, but rather than possibly serving as a mechanical implant, as is possible with the sixth embodiment, the mechanism 104 can be used to inject a fluid such as a hydrogel, PMMA bone cement, BMP, silicone, or the like, into the facet joint 36. The distraction mechanism 104 can be mounted on the distal end of a delivery tool 114, which uses a thumb slide 118 for applying energy to the mechanism 104. The distraction mechanism 104 comprises a somewhat flattened and rigid blade 120 that is slightly S-shaped in cross-section and is hollow with the hollow interior of the mechanism 104 communicating with lateral ports 122 in notches 124 formed in the side edges of the blade 120. When the mechanism 104 is inserted into a facet joint 36, it is inserted with its relative flat dimension oriented generally parallel to the articular surfaces of the joint 36. To obtain distraction, the mechanism 104 can be rotated with the delivery tool 119, which can cause the joint 36 to distract. Once the joint 36 has been distracted, a gel or other suitable fluid can be emitted through manipulation of the thumb slide 118 on the delivery tool 119 so that the gel is forced through the hollow interior of the device and out the injection ports 122 to fill the facet joint 36 to the desired level. The distraction mechanism 104 can then be withdrawn with the gel serving as an implant to retain the distracted joint 36 at the desired separation.

As with the rest of the embodiments of the invention described herein, in some versions of the embodiment depicted in FIGS. 12A-12D, the tool 119 may include a central delivery lumen through which an implant may be delivered, for example, via a plunger or push rod, to the distracted facet or joint space. In other versions of the embodiment depicted in FIGS. 12A-12D, another delivery tool may be employed once the distraction mechanism 104 has distracted the joint 36.

An eighth embodiment of the invention is shown in FIGS. 13A-13D. In this embodiment, the distraction mechanism 126 can have a generally hollow elongated body 130. In some embodiments, the body 130 may be relatively cone-shaped. The thickness of the body 130 can be small enough so it can be initially inserted a short distance into a relatively flat and collapsed facet joint 36. The body 130 can include facet engaging features 132 including diametrically opposed deployable keels (FIG. 13B), thread slots (FIGS. 13A and 130), or deployable teeth (FIGS. 13E, 13F, and 13H). The keels can have a retracted position wherein they lie within the confines of the body 130, but, when activated as indicated with the arrows in FIG. 13B, extend in a tapered manner away from the outer surface of the body 130. The thread slots shown in FIG. 13A can be adapted to receive threads from an actuator that can be advanced through the body 130 to distract the facet joint 36. The threads of the actuator can gain purchase in the articular surfaces of the facet by protruding through the thread slots of the body 130. The deployable teeth, like the keels, can have a retracted position wherein they lie within the confines of the body 130, but, when activated, extend through openings in the body 130 to engage the articular surfaces of the facet joint.

The distraction mechanism 126 can include an actuator as shown in the form of an elongate member 134. The elongate member 134 can be a threaded member or a toothed ratchet type member adapted to be inserted through the hollow core of the body 130. As shown in FIGS. 13A and 138, the

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elongate member 134 can be a screw or bolt type member adapted to threadably engage the body 130. The elongate member 134 can be actuated and advanced via distraction energy applied to the tool 129 via a handle 128 as previously described. In the case of a body 130 having keels, the advancing elongate member 134 can result in the expanding keels engaging the facet surfaces of the joint 36 causing separation of the joint 36 as increasingly more distraction energy is applied resulting in progressively more deployment and height of the device with the keels also providing fixation to the facet surfaces. In the case of threaded slots, the advancement of the elongate member 134 can expand or stretch the body 130 or merely advance through the body 130. As the threads of the elongate member 134 engage the threaded slots, the threads can protrude through the slots and cut into or otherwise engage the articular surfaces of the facet joint 36. In the case of deployable teeth, the advancing elongate member 134 can force the teeth through openings in the body 130 thereby causing them to engage the facet surfaces of the joint 36 allowing for secured positioning of the implant.

As shown, the elongate member 134 can include an engagement feature 135 for receiving a corresponding engagement feature from the cannula 127 of the delivery tool. As shown the engagement feature 135 of the elongate member can be adapted to receive a straight screw driver tip. Other rotating and/or driving engagements can be provided.

A ninth embodiment of the invention is shown in FIGS. 14A-14C. In this embodiment, the distraction mechanism 138 may include one or more leaf blades 142 with tapered tips 144. The blades 142 may extend longitudinally from a delivery tool cannula 137 and may be stacked relative to one another. The blades 142 may be actuatable independent of one another and may be actuatable in consecutive order.

The blades 142 can be made of steel, titanium, PEEK, carbon, or nitinol. Other materials can be used. The blades 142 can be configured as temporary distraction shims or permanent detachable implants which can be left in the distracted joint 36. If the stacked blades 142 are configured for detachment and permanent implantation, they can include one or more of the previously described fixation mechanisms. Additionally, the blades 142 can include relative motion restraints adapted to prevent relative slippage from occurring between blades 142. These restraints can include detent recesses and protrusions or other restraints.

The distraction achieved with this system can occur by advancing the blades 142 one at a time into the facet joint 36 so that as an increasing thickness or height of the stack of blades 142 is inserted into the joint 36, distraction is achieved. The number of blades 142 extended into the joint 36 can be increased until the desired distraction of the joint 36 is achieved.

The handle 140 of the present embodiment can include any of the handles previously described. However, as shown in FIGS. 14B and 14C, the handle 140 can include a rotating dial type handle 140 such that rotation of the dial relative to the handle 140 causes actuation of the blades 142. This can occur through an internal screw system that can convert radial motion to longitudinal motion. Additionally, the proximal ends of the blades can be initially staggered such that an internally advancing rod can first engage and advance the bottom blade, and then the next blade, and the next blade, and so on. Accommodations can also be made to avoid continued advancement of a given blade 142 once it enters the joint. Accordingly, incremental rotation of the dial can cause incremental advancement of the internal rod and thus consecutive advancement of blades 142.

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The dial can include measurements corresponding to the thickness of each of the advancing blades **142** and can thus display to the user, the total distraction being provided by the blades **142**. That is, rotation of the dial can advance a first blade **142** into the joint **36** and the indication on the dial can reflect the thickness of the first blade **142** and thus the resulting distraction. Upon further rotation of the dial, additional blades **142** can be advanced into the joint **36** and the indication on the dial can reflect the cumulative thickness of the first blade **142** and the additional blades thereby reflecting the total distraction.

Accordingly, the present embodiment can be used to distract a facet joint **36** as described and can also be used as an implant. However, the tool **139** can also be used to calibrate or prepare for a distraction procedure by assisting a user in determining the appropriate amount of distraction. In one embodiment, the tool **139** described can be used to obtain intra operative feedback from a patient. The tool **139** can be used to incrementally distract a joint **36** to determine how much distraction is necessary to alleviate patient symptoms. Based on feedback from the patient, an appropriately sized implant can be selected for insertion into the facet joint **36**. The current embodiment can then be used to maintain the facet joint **36** in a distracted condition for placement of the implant or other devices can be used to place the implant.

In one version of the embodiment depicted in FIGS. **14A-14C**, each of the blades **142** of the mechanism **138** may have a longitudinally extending slot (not shown) defined therein and positionally corresponding with the slots of the adjacent blades **142** to define an overall slot (not shown) through which an implant may be delivered, for example, via a plunger or push rod, to the distracted facet or joint **36** space. In other versions of the embodiment depicted in FIGS. **14A-14C**, another delivery device may be employed once the mechanism **138** is used to distract the facet joint **36**.

Referring to FIGS. **15A-15C**, a tenth embodiment of a distraction mechanism **148** of the invention is illustrated. In this embodiment, a rotatable loop **148** of rectangular configuration made of a rigid material can be pivotally mounted on the distal end of the delivery tool **149** and can carry within the loop **148** an implant of the type disclosed in the prior embodiments, or of types to be disclosed hereafter, which can be deposited or left in a distracted joint **36**. The loop **1478** can have elongated side legs **154** and relatively short end legs **156** with fixation mechanisms **158** such as teeth on both surfaces of the short legs **156** of the loop **148**. The pivotal mount between the loop **148** and the delivery tool **149** can be positioned approximately midway along the length of the side legs **154**.

Distraction energy can be applied to the rigid loop **148** from the handle end of the delivery tool **149** causing the trailing end of the loop **148** to rotate forwardly, as illustrated with the arrows in FIG. **15A** (posterior to anterior). Together with the teeth on the surface of the loop **148**, the loop **148** can engage the facet surfaces to separate the joint **36**. The forward rotation of the loop **148** can distract the joint **36** and also push the superior facet slightly forward in translation relative to the inferior facet.

When the desired distraction and translation of the facet joint **36** is achieved, the tool **149** can deposit the implant, which is carried within the loop **148**. The implant can be carried within a hollow inside the cannula **147**, for example. The implant can be introduced into the distracted joint **36** and deposited therein prior to the mechanism **146** being removed.

An eleventh embodiment of a distraction mechanism **160** of the present invention is shown in FIGS. **16A-16C**. The

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distraction mechanism **160** can include an expandable receiving portion having upper **164** and lower **166** generally planar elements which are connected at an end with a living hinge so that the mechanism **160** can be flattened for insertion into a facet joint **36**. As shown, the upper and lower generally planar elements **164**, **166** can be relatively triangularly shaped or trapezoidally shaped with the living hinge being positioned near the narrower end of the respective shapes. Other shapes, such as, for example, rectangular, of the planar elements **164**, **166** can be used. The outer surface of the planar elements **164**, **166** can include tapered keels as fixation mechanisms **170** which assist in gripping the superior and inferior facet surfaces. The keels can be provided along the edges of the planar surfaces **164**, **166** as shown or they can be provided more centrally to the planar surfaces **164**, **166**. The outer surface can also include threaded slots as fixation mechanisms **170** for receiving threads from an actuator being advanced into the receiving portion. The outer surface can also include teeth as fixation mechanisms **170** adapted to engage the facet surfaces.

An actuation device **172** can be included, as shown in FIG. **16A**, and can take the form of a wedge, an elongated member, or block-type member. In the case of a wedge actuation device, the wedge can be advanced into the receiving portion between the planar members **164**, **166** forcing the planar members **164**, **166** apart and distracting the joint **36**, as shown in FIG. **16B**. The fixation mechanisms **170** on the surface of the planar members **164**, **166** can engage the facet surfaces thereby resisting any tendency for the mechanism **160** to back out of the joint **36**.

In the case of an elongate member, the elongate member can be advanced between the planar members **164**, **166** forcing the planar members **164**, **166** apart and distracting the joint **36**. As shown, the elongate member may include a tapered tip to initiate the advancement. In addition, the elongate member can include threads to facilitate advancement into the receiving portion. Moreover, where threaded slots are provided on the planar members **164**, **166**, the threads of the elongate member can engage the threaded slots and protrude through the threaded slots. Accordingly, the threads can cut into or otherwise gain purchase in the facet surfaces to prevent any back out tendency of the mechanism **160**.

In the case of a block-type actuation device, the block can be advanced between the planar members **164**, **166** forcing the planar members **164**, **166** apart and distracting the joint **36**. The block can include a taper (not shown) at a distal end for initially engaging and separating the planar members **164**, **166**. Alternatively or additionally, a portion of the planar members **164**, **166** may extend outside the facet joint **36** and may be pried apart for initial insertion of the block after which the forced advancement of the block can cause separation of the facet allowing the block to be fully inserted into the receiving portion and into the facet joint **36**. Where the receiving portion includes teeth and/or keels, these fixation mechanisms **170** can be forced into the facet surfaces gaining purchase therein.

The receiving portion of the mechanism **160** can be carried on the distal end of a delivery tool **159** and can be inserted into the facet joint **36** in a collapsed state. One of the actuation devices **172** described can then be inserted into the receiving portion from its open trailing end to expand the receiving portion by separating the planar elements **164**, **166** until the desired distraction of the joint **36** is achieved. This separation can lead to distraction and forward translation of the facet joint **36**.

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As shown in FIGS. 16A and 16B, the receiving portion can conform to the shape of the articular surfaces of the facet joint 36 as described with respect to the embodiment shown in FIG. 9. As such, the distraction mechanism 160 can distribute the compressive load from the joint 36 more evenly over the surface interacting with the distraction mechanism 160. This shape conformance can also function to resist withdrawal or dislodgement of the mechanism 160 or implant. It is further noted that this shape conformance can be multidirectional and where the keels are positioned along the edges of the planar elements 164, 166, the keels may facilitate folding or bending of the planar surface 164, 166 around the actuation device 172 allowing a fixation feature more central to the planar surface 164, 166 (e.g., teeth) to also engage the facet surface.

The distraction mechanism 160 in this embodiment can be removably or permanently mounted on the distal end of an insertion tool 159 so it can be removed from the insertion tool 159 and left in the distracted joint 36 as a permanent implant or removed from the distracted joint 36 after an auxiliary implant is positioned therein.

With reference to FIGS. 17A-17C, a twelfth embodiment of the invention is illustrated. As best appreciated by reference to FIGS. 17A and 178, the distraction mechanism 174 can be configured like a medical capsule (FIG. 17B) so as to be cylindrical in configuration with rounded leading and trailing ends 176. The mechanism 174 can be formed from two generally ovalar or tear-drop shaped components with an upper component 182 having fixation mechanisms 180 on its upper surface and the lower component 186 having fixation mechanisms 180 on its lower surface. The components can be slidably related along a diagonal plane 188 of separation.

In use, the mechanism 174 can be inserted into the facet joint 36 with a delivery tool 169 having the mechanism 174 positioned on its distal end. As the mechanism 174 is advanced, the fixation elements 184 can grip the opposing facet surfaces causing the tear-drop components 182, 186 to translate along the diagonal plane 188 in one direction or the other. In one direction, of course, the implant becomes longer with a maximum height of the diameter of the capsule as seen in FIG. 17B while in an opposite direction the components 182, 186 slidably converge along the diagonal plane 188 so that the implant is cammed into a taller or thicker dimension causing more distraction of the joint 36. The mechanism 174 can have any number or type of fixation mechanisms 180 on the top and bottom surfaces to prevent migration.

The opposing surfaces along the diagonal plane 188 can be somewhat smooth to prevent excessive friction while facilitating small amounts of motion along the diagonal plane 188 once the mechanism 174 is permanently fixed against the superior and inferior facet surfaces.

The mechanism 174 can be made of steel, titanium, PEEK, silicone, plastic, polymer, or nitinol. The mechanism 174 can be detachable to facilitate permanent implantation, but, as with the other embodiments, can be removed once an auxiliary implant is positioned upon distraction of the joint 36 with the mechanism 174.

A thirteenth embodiment of a distraction mechanism 192 of the invention is shown in FIGS. 18A and 181. The mechanism 192 of this embodiment can include a polyethylene balloon 194 mounted at the distal end of a delivery tool. The distal end of the delivery tool within the balloon 194 can include a large conduit 198 with two separated enclosed vessels 200, 202 and an output port 204 on the bottom of the device communicating with an inlet vessel 200

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and an intake port 206 on the top of the device communicating with a return vessel 202. Fluid can be injected into the balloon 194 through the inlet vessel 200 and removed as desired through the return vessel 202 as shown by the directional arrows in FIGS. 18A and 188.

In use, the balloon 194 can be positioned within the facet joint 36 and the balloon 194 can be inflated through the introduction of a contrast medium fluid. Pressure within the balloon 194 can be developed through further injection of fluid and thus the pressure can cause expansion of the balloon 194 structure. The balloon 194 expansion can cause separation of the facet joint 36 resulting in distraction and forward translation of the joint 36.

Once in place and expanded, as with other distraction mechanisms, the mechanism 192 can be detached from the tool and used as an implant or the mechanism 192 can be used to hold the distraction of the joint 36 while another implant is placed. Where used as implant, the balloon 194 can be filled with a bio-inert hydrogel once optimal distraction and translation of the facet joint 36 is achieved. Also, a sealing valve at the proximal aspect of the balloon 194 can be provided to prevent leaking of the hydrogel. Alternatively or additionally, the balloon 194 can be inflated with a fast-curing silicone when used as a permanent implant.

In another embodiment of the balloon 194 (not shown), the balloon 194 can be forked shaped having two legs and a recess therebetween in which a permanent implant of any of the types previously described, or to be later described, can be positioned. In this configuration, the balloon 194 can be inflated to achieve optimal distraction and translation, but can be removed following the successful placement and fixation of the permanent implant.

A fourteenth embodiment of a distraction mechanism 208 of the invention is shown in FIGS. 19A and 196 and operates similarly to that of the thirteenth embodiment shown in FIGS. 18A and 18B. In the fourteenth embodiment, the main conduit 210 on the distal end of the delivery tool can have a large enclosed channel 214 formed therein and a smaller enclosed channel 216 communicating with the large channel. The large channel 214 can have an extended cylindrical vessel 218 protruding beyond the end of the main conduit and a rounded tip 220 with a lateral outlet port 222 in a recess 224 formed in a side thereof. A balloon 226 can be sealed to the large vessel 210 so that fluids emitted through the outlet port 222 can fill and expand the balloon 226 to distract the facet joint 36 as with the thirteenth embodiment, and the smaller enclosed channel 216 can be used to remove fluid from the balloon 226 as desired.

A fifteenth embodiment of the distraction mechanism 228 of the invention is shown in FIGS. 20A-20C. The mechanism 228 can again be mounted on the distal end of a delivery tool with the mechanism 228 being a hollow cylinder 232 having tapered diametrically opposed protruding prongs 234 at its leading end. The prongs 234 can define a recess 236 therebetween and a hollow interior of the device can communicate with an injection port 238 positioned between the prongs 234. The tapered prongs 234 can be advanced into a facet joint 36 to spread the joint 36 apart to allow room to inject fluid through the injection port 238. The injection port 238 can be used for communicating a stylet to the facet joint 36 or for introducing materials (including bone graft, BMP, OP1, silicone, PMMA bone cement, or hydrogel) and permanent implants to provide permanent distraction and translation.

A sixteenth embodiment of the distraction mechanism 240 of the invention is shown in FIGS. 21A-21C with this mechanism 240 again being mounted on a delivery tool. The

mechanism **240** can include a pair of parallel tubular members **244** on lateral aspects of the distal tip of the delivery tool and a port **246** between the tubular members **244** communicating with the hollow interior of the delivery tool. Each tubular member **244** can have a relatively small leading end but become progressively larger as bigger tubes are telescoped over the smaller tubes. The larger tubes can cause increasing separation of the facet joint **36** resulting in distraction and forward translation. The port can facilitate the introduction of subsequent instruments, stylets, materials (including bone graft, BMP, OP1, silicone, PMMA bone cement, hydrogel) and permanent implants to provide permanent distraction and translation.

A seventeenth embodiment of the invention is shown in FIGS. **22A-22D**. In this embodiment, the distraction mechanism **248** is mounted on the tip of a delivery tool having dual triggers **252, 254** for delivering energy to the mechanism **248**. The mechanism **248**, as seen best in FIGS. **22B-22D**, can include upper and lower arms **256, 258** having confronting flat faces with at least the upper arm being somewhat flexible while both arms are substantially rigid. The lower arm **258** can have a rearwardly and upwardly beveled leading tip **260** and a hollow interior in communication with side ports **262** on opposite sides of the arm **258**. The upper arm **256** can have a relatively flat main body which overlies in confronting relationship the lower arm **258** with the upper arm **256** having a downwardly and forwardly inclined leg **264** which conforms with the bevel of the lower arm **258**. When the upper and lower arms **256, 258** are collapsed, as shown in FIG. **22B**, they lie in confronting contiguous relationship with the leading edge of the upper arm **256** being close to the elevation of the lower edge of the lower arm **258** so as to provide a thin profile for insertion into a facet joint **36**. One trigger on the delivery tool can retract the upper arm **256** so that the inclined leg **264** at the leading end thereof is cammed upwardly onto the top of the lower arm **258** by the leading beveled edge **260** of the lower arm **258**. The retraction of the upper arm **256** can separate the upper arm **256** from the lower arm **258** as shown best in FIGS. **22C** and **22D** so that when they are positioned within the facet joint **36**, a distraction of the joint **36** can be achieved. Once the joint **36** is distracted to the desired degree, the introduction of instruments, stylets, materials (including bone graft, BMP, OP1, silicone, PMMA bone cement, hydrogel) or other permanent implants of the type previously disclosed herein can be introduced through the side ports **262** with the second trigger to provide permanent distraction and translation.

An eighteenth embodiment of the distraction mechanism of the invention is shown in FIGS. **23A-23D**. In this embodiment of the invention, the distraction mechanism **266** can again be mounted on the distal end of a delivery tool having dual triggers **270, 272**. The mechanism **266** can have upper and lower blades **274, 276** each having forwardly tapered leading ends **278** so as to define a relatively thin leading tip resembling an elongated duckbill. The tip of the duckbill for the mechanism **266** can be thin enough or flat enough to access the facet joint **36**. Once the mechanism **266** is positioned within the facet joint **36**, a distraction energy can be applied to the mechanism **266** by pulling one trigger which causes an internal hollow cylinder **280** having a blunt leading end to advance forwardly thereby separating the upper and lower blades **274, 276**. The separation can cause distraction of the facet joint **36** and lateral injection ports **282** can be provided on the internal cylinder through which instruments, stylets, materials (including bone graft, BMP, OP1, silicone, PMMA bone cement, or hydrogel) can be

injected into the facet joint **36** by the second trigger or other permanent implants could be introduced to provide permanent distraction and translation.

A nineteenth embodiment of the invention is shown in FIGS. **24A-24D** to be similar to the eighteenth embodiment. In the nineteenth embodiment, again, upper and lower relatively rigid yet flexible blades **286, 288** can be provided that in a closed position are contiguous and confronting with each other. Adjacent the leading end **298** of the blades **286, 288** the lower blade **288** can have a transverse cylindrical recess **290** formed therein while the upper blade **286** can have a transverse cylindrical protrusion **294** which seats in the recess **290** to permit the upper and lower blades **286, 288** to be contiguous in the closed position. An internal cylinder **294** having lateral ports **296** at its leading end can be positioned between the blades **286, 288** and confined therebetween when the blades **286, 288** are closed so that tapered forward ends **298** of the blade can be inserted into the facet joint **36**. By pulling the first trigger on the delivery tool, the upper blade **286** can be retracted rearwardly so that the cylindrical protrusion **292** is cammed by the cylindrical recess **290** to cause the upper blade **286** to elevate relative to the lower blade **288** thereby distracting the facet joint **36**. When the facet joint **36** has been distracted, instruments, stylets, materials (including bone graft, BMP, OP1, silicone, PMMA bone cement, or hydrogel) can be injected into the facet joint **36** with the second trigger or other permanent implants of the types previously disclosed herein could be introduced to provide permanent distraction and translation.

FIGS. **25A-25F** disclose various embodiments of permanent implants which can be inserted into a facet joint **36**, for example, with a delivery tool of the type disclosed in the tenth embodiment or any other of the embodiments disclosed herein if such embodiments are supplemented with an integral delivery lumen and push rod configuration. Alternatively, the various embodiments of the above-described distraction mechanisms can be positioned in a facet joint **36** to retain the joint **36** in a distracted position, a translated position, or a combination distracted and translated position until a permanent implant can be positioned in the distracted joint **36** separately via a separate delivery device or tool.

FIG. **25A** shows a permanent implant **306** that is of wedge shape having fixation mechanisms **308** on the top and bottom surfaces comprised of teeth all directed in a common direction toward the trailing end of the implant. The wedge shape can facilitate distraction that is both normal to the facet surfaces and parallel to the facet surfaces. In other words, the wedge shape can separate the facet surfaces from each other in a direction that is perpendicular and a direction that is parallel to the facet surfaces, thus both increasing the offset distance between the facet surfaces and translating the facet surfaces relative to each other. This combination of displacement can more fully open the foramina spaces.

FIG. **25B** shows another implant **310** having rounded leading and trailing ends **312** and flat upper and lower parallel surfaces **314**, again with teeth on both the upper and lower surfaces **314**, which are inclined rearwardly.

FIG. **25C** discloses another wedge-shaped embodiment of a permanent implant **318** where the upper surface **320** of the wedge has teeth **322** directed forwardly and outwardly along opposite lateral edges at the leading edge of the wedge while the bottom surface **324** has similar teeth **322** at the trailing end of the wedge directed rearwardly and outwardly. The oppositely oriented teeth **322** may enhance and help to maintain the translation displacement already provided by the wedge shape.

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FIG. 25D shows a permanent implant 326 configured similarly to FIG. 25B with the trailing end 328 being squared off rather than rounded and with teeth 330 across the top of the leading end directed forwardly and teeth 330 across the bottom and the trailing end directed rearwardly. The oppositely oriented teeth 330 may provide a translation displacement.

FIG. 25E shows still another implant 332 of generally rectangular configuration but having a concave leading end 334 and a rounded trailing end 336. The implant 332 has rearwardly directed fixation mechanisms 338 in the form of teeth along the upper surface on lateral sides and along the side walls.

Permanent implants can vary in geometry, material, and fixation mechanism. For example with respect to geometry a wedge shaped implant can provide for a greater height of the posterior aspect of the implant relative to the anterior aspect of the implant. The wedge can also provide for uniform dimensions at the lateral and medial aspects of the implant. The wedge shape may result in a translating vector force and a separating vector force that results in both subluxation and distraction, thereby increasing the foramina space more fully as discussed in U.S. provisional patent application No. 61/059,723, filed Jun. 6, 2008, and incorporated by reference herein in its entirety. A double wedged implant can provide greater height of the posterior aspect of the implant relative to the anterior aspect of the implant in addition to greater height of the lateral aspect of the implant relative to the medial aspect of the implant (see FIG. 25F). Other geometrical variations can include a flat rectangular shape, an oval pill shape, a concave superior surface, a concave inferior surface, a convex superior surface, a convex inferior surface, a convex anterior surface, a concave anterior surface, a convex posterior surface, and a concave posterior surface.

With respect to materials, several materials can be provided including steel, PEEK, carbon, allograft, polymer, and silicone. With respect to fixation mechanisms, at least three mechanisms can be included. Aggressive shark teeth can be provided with a directional orientation positioned to achieve optimal fixation relative to the natural biomechanics of various sections of the spine. The teeth can be long enough to gain purchase in the cortical bone of the facet surfaces. Cleats can also be provided that have a less aggressive profile than the shark teeth but still allow for directional orientation for the same reasons described above. These cleats can also be capable of anchoring in the hard cortical bone of the facet surface. Additionally, a roughened pore surface can be provided to prevent free sliding of the implant within the facet joint 36. These surfaces can be roughened and coated with commercially available resurface chemicals that would create friction and prevent device migration.

Any or all of the implants can be adapted as fusion type implants or motion preservation type devices. Implants with varying degrees of motion preservation can also be provided. In the case of a motion preservation type implant, the implant can have fixation mechanisms on one side to enable both temporary and permanent fixation to one surface of the facet joint 36 while allowing the opposing facet surface to slide freely across the surface of the implant. The facet joint 36 can be a naturally sliding joint 36 and a distraction implant with fixation on only one side may accommodate the natural sliding of the facet. However, in some circumstances, a fusion type implant can be more suitable. In these circumstances the implant can include fixation mechanisms on both sides of the of the implant.

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FIGS. 26A-26C disclose a twentieth embodiment of a distraction mechanism of the invention which is somewhat similar to the embodiments of FIGS. 22A-D and 23A-D. In this embodiment of the invention, upper and lower confronting bars 342, 344, which are relatively rigid but have flexibility, can be retained in a delivery tool so that the leading tip 348 of the distraction mechanism 340 is of a duckbill structure with the leading tip 348 of both the upper and lower bars 342, 344 being beveled to form a thin leading end of the mechanism 340 which can be inserted into the facet joint 36. The upper and lower bars 342, 344 can define a longitudinal channel therebetween in which a plunger 350 can be slidably disposed and reciprocally moved with a finger grip 352 on the delivery tool. Forwardly of the plunger 350, a permanent implant 354 can be positioned within the channel between the upper and lower bars 342, 344 so that upon forward movement of the plunger 350, the implant 354 can spread the upper and lower bars 342, 344 allowing the implant 354 to be released from the leading end 348 of the distraction mechanism 340 and be deposited in the facet joint 36. The upper and lower bars 342, 344 can be biased toward each other by an elastic band 356 which passes around the upper and lower bars 342, 344 forwardly of the delivery tool but rearwardly of the leading tip 348 of the distraction mechanism 340.

Although the present invention has been described with a certain degree of particularity, it is understood the disclosure has been made by way of example, and changes in detail or structure may be made without departing from the spirit of the invention as defined in the appended claims. In particular, it is noted that several embodiments have been described with particular features. It is to be understood that the features of any given embodiment can be combined with features of other embodiments and still be within the scope of the invention. For example, the elongate member of FIGS. 9A-9C has been described as a toothed member adapted for a ratcheting action, while in other embodiments the elongate member has been described as a threaded member. Where functionality allows, interchanging certain features of one embodiment with another embodiment is within the scope of the present invention.

What is claimed is:

1. A method for distracting a facet joint formed by two adjacent vertebrae of a spine of patient, the method comprising:
 - advancing a facet distraction implant into the patient and toward the facet joint;
 - delivering, via a tubular delivery device, the facet distraction implant into the facet joint, wherein the facet distraction implant comprises a top surface, a bottom surface, a proximal end face portion opposite a distal portion and multiple fixation features disposed along the top and bottom surfaces
 - wherein the delivery device comprises a distal end portion directly engaged with and releasably coupled with at least a portion of the proximal end face portion of the implant.
2. The method of claim 1, wherein the top surface and the bottom surface comprise malleable material that conforms to the facet joint upon implantation.
3. The method of claim 1, wherein the top surface is a bone facing surface configured to engage a superior aspect of the facet joint upon implantation.
4. The method of claim 1, wherein the bottom surface is a bone facing surface configured to engage an inferior aspect of the facet joint upon implantation.

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5. The method of claim 1, wherein the fixation features are bone contacting features.

6. The method of claim 1, wherein the bone contacting features are teeth.

7. The method of claim 1, wherein the facet joint is the cervical facet joint.

8. The method of claim 1, wherein the advancing further comprises advancing from a posterior direction relative to the spine of the patient.

9. A method for distracting a facet joint formed by two adjacent vertebrae, the method comprising:

advancing a facet distraction implant toward the facet joint;

delivering, via a tubular delivery device, the facet distraction implant into the facet joint, wherein the facet distraction implant comprises a proximal end face portion, a distal portion, a top surface, a bottom surface, and each of the top and bottom surface comprises one or more bone contacting features,

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wherein the delivery device comprises a distal portion directly engaged with and releasably coupled with at least a portion of the proximal end face portion of the implant.

10. The method of claim 9, wherein the top surface and the bottom surface comprise malleable material that conforms to the facet joint upon implantation.

11. The method of claim 9, wherein the top surface is a bone facing surface configured to engage a superior aspect of the facet joint upon implantation.

12. The method of claim 11, wherein the bottom surface is a bone facing surface configured to engage an inferior aspect of the facet joint upon implantation.

13. The method of claim 9, wherein the bone contacting features are teeth.

14. The method of claim 9, wherein the facet joint is the cervical facet joint.

15. The method of claim 9, wherein the advancing further comprises advancing from a posterior to anterior direction.

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